

From Unstructured Data to Demand Counterfactuals: Theory and Practice^{*}

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Abstract

Empirical models of demand for differentiated products rely on low-dimensional product representations to capture substitution patterns. Practitioners increasingly generate proxies for the true representations from high-dimensional, unstructured data, including product descriptions, reviews, and images. When proxies fail to capture the true dimensions of product differentiation, standard workflows deliver biased counterfactuals and invalid inference. We develop a practical toolkit that corrects this bias and ensures valid inference for counterfactuals. Our methods apply to market-level and/or individual data, require minimal additional computation, are efficient, deliver simple formulas for standard errors, and accommodate data-dependent proxies generated from fine-tuned ML models. They can also be used when standard quantitative attributes are imperfect proxies. Further, we propose diagnostics to assess the adequacy of the proxy construction and dimension. Our methods yield meaningful improvements in predicting counterfactual substitution in both simulations and an empirical application where second choice data, excluded from estimation, provide a ground truth to assess counterfactual prediction performance.

Keywords: Unstructured data, demand estimation, bias correction, fine tuning, product embeddings, differentiated products.

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1 Introduction

Estimating demand for differentiated products is central to many fields of economics and marketing. A common strategy is to specify the utility of a product as a function of the product’s price and other observable attributes, often allowing for rich types of consumer heterogeneity (Berry et al. (1995, 2004), henceforth BLP). Among other applications, this approach has been used to study the impact of horizontal mergers (Nevo, 2000a), new product launches (Hausman, 1994; Petrin, 2002), trade policy (Goldberg, 1995), school choice (Bayer et al., 2007; Neilson, 2017), two-sided markets (Fan, 2013; Lee, 2013), the evolution of markups over time (Grieco et al., 2024), and how firms should optimize prices and promotions (Vilcassim and Chintagunta, 1995).

The success of demand models in predicting counterfactual quantities of interest (*counterfactuals* hereafter) hinges on their ability to capture substitution patterns. Doing so requires knowledge of the product attributes that correctly reflect the underlying dimensions of differentiation, which poses a fundamental measurement challenge (Berry and Haile, 2021). First, consumer choices are often driven by hard-to-quantify characteristics, such as visual design, user friendliness, or style. In these cases, a growing literature shows that product images, descriptions, and reviews contain valuable information to capture substitution (Compiani et al., 2025; Han and Lee, 2025; Lee, 2025; Caoui, 2025). Consumer surveys and product recommendations on platforms may also provide useful measures of product differentiation (Magnolfi et al., 2025; McClure, 2025). To use these high-dimensional, unstructured data in demand models, practitioners transform them into lower-dimensional numerical variables, or *embeddings*, using machine learning (ML) methods. Second, even numeric attributes could be imperfect proxies (e.g., Nevo, 2001; Allcott and Wozny, 2014), or could be high-dimensional and collinear, requiring dimension-reduction (e.g., Backus et al., 2021). In all of these cases, the variables used as inputs in the demand model are *proxies* for the true attributes that drive consumer choices. It is essential that these proxies adequately capture the true dimensions of differentiation: poor proxies can lead to biased estimates of demand model parameters and biased counterfactuals.

In this paper, we propose a simple, post-estimation *bias correction* for counterfactuals. We take the *naive estimator* that treats the proxies as if they were the true dimensions of differentiation (as is implicitly done in practice) and add a correction term designed to achieve two goals. First, it removes the asymptotic bias arising

from using imperfect proxies. Second, it makes the bias-corrected estimator *efficient*, meaning it has the smallest possible asymptotic variance when proxies may be imperfect. This is particularly important as it is well known that estimating substitution can be data demanding. The intuition for the bias correction is simple: when proxies are imperfect, the model fit will tend to suffer in a way that correlates with the bias in the counterfactual estimator. Our approach leverages this correlation, mapping the discrepancy between the model and the data into a correction for the counterfactual. We also provide closed-form standard errors, making it easy to perform valid inference on counterfactuals. In addition, we show how two simple *diagnostics* can be used to assess the adequacy of different proxies in capturing substitution. These diagnostics help guide the choice of *how many* and *which* proxies/attributes should be included—questions that practitioners need to answer in any instance.

We develop these methods for two widely-used empirical frameworks. The first follows [Berry et al. \(1995, 2004\)](#): prices vary across markets, instrumental variables are used to address price endogeneity, and market-level data may be supplemented with individual choice data. Many papers in industrial organization (IO) and fields using IO tools fit in this category. The second consists of models estimated on individual choice data with product-level fixed effects, which are common in marketing applications (see [Dubé and Rossi \(2019\)](#) for a review). While we illustrate our approach for workhorse specifications (e.g., mixed logit with normal random coefficients), the methods apply to a broad class of parametric functional forms.

The bias corrections and diagnostics are computationally light and integrate easily into the standard demand estimation workflow. The bias corrections take the naive parameter estimates produced by standard packages (e.g., PyBLP or xlogit) as inputs, and require neither bootstrapping nor additional optimization. Similarly, the diagnostics are Lagrange Multiplier (LM) statistics evaluated at the naive estimates.

Our methods are based on two key insights. The first is that the demand model is misspecified when proxies are imperfect. Importantly, *this is not a measurement error problem*: the proxies are at the product level whereas the unit of observation is the market and/or individual. By contrast, mismeasurement is at the observation level in a standard measurement error problem. The second insight is to reparameterize the model with a composite parameter that captures how proxies interact with structural parameters to affect utilities. The benefit of this framing is that it allows us to then address the misspecification problem using standard two-step methods.

A further advantage of this approach is that it allows us to be agnostic about the nature of the proxy error. This is particularly valuable because proxies are often obtained via black-box ML models, making it difficult to justify specific assumptions on the error. This approach also accommodates data-dependent proxies, such as those obtained by *fine tuning*. For instance, practitioners may update the embeddings produced by off-the-shelf LLMs or neural networks to better fit the choice data. Moreover, this approach does not require taking a stand on the units of the proxies and/or true dimensions of differentiation. This is especially important for hard-to-quantify characteristics like visual design or user friendliness that lack natural units.

Two simulations calibrated to a standard data set (Nevo, 2001) and to the empirical application below confirm that the bias correction improves performance for a range of levels of proxy error. Specifically, when bias in the naive estimator is material, the corrected estimator has lower bias and lower RMSE. Our formulas for standard errors also deliver confidence intervals that exhibit good coverage of the true counterfactual even when the naive estimator is substantially biased.

Finally, using the experimental data from Compiani et al. (2025) (CMS hereafter), we show that the bias correction meaningfully improves the model’s ability to predict counterfactual choices following product removals. To this end, we leverage the fact that the data features both consumers’ first and second choices. We estimate model parameters on the first choice data, then compare how well the naive and bias-corrected estimators predict second choices. This provides a ground truth to validate the performance of our approach in predicting counterfactuals. The bias correction improves the model’s ability to predict a product’s closest substitute from 40% (with the naive estimator) to 70%. Further, our diagnostics correctly identify the set of proxies that perform best at the counterfactual prediction task, indicating that they can be valuable tools for practitioners.

The mechanism behind the improvement is informative. The predictions are based on a mixed logit model with random coefficients on the embeddings. As documented in our simulations, when embeddings are poor proxies they are effectively discarded and the naive predictions revert to those of a plain logit model. This is what we see in the application: the naive predictions are biased in the direction of a plain logit model, whereas the bias correction recovers the true second-choice behavior.

Our approach is also helpful for practitioners using standard numeric attributes, as proxy error may be present even in this case, especially when dimension-reduction

methods (e.g., PCA) are used to shrink the attribute set. Further, the choice of which numeric attributes to include is generally ad hoc, and our diagnostics can help practitioners in making these decisions. Whether or not proxy error is a concern, an additional contribution of this paper is to provide easy-to-compute, efficient estimators and standard errors for counterfactuals across many empirical settings, including combined market-level and microdata (e.g., [Petrin, 2002](#); [Berry et al., 2004](#), and many subsequent works). To the best of our knowledge, these contributions are new.¹

Our approach is related to double/debiased ML (DML).² Both approaches aim to estimate a target parameter in the presence of nuisance parameters. In our setting, the target is the counterfactual and the nuisances are both the latent dimensions of differentiation, which are “estimated” using proxies, and the demand model parameters, which are themselves estimated using the proxies as inputs. This multi-step dependence on the proxies combined with the fact that the proxies are often outputs of black-box ML models trained on data to which one might have no access makes our setting quite different. Nevertheless, both approaches rely on standard orthogonalization ideas from the semiparametrics literature.

A recent literature recognizes that naively treating ML-generated variables as data leads to measurement-error bias and develops corrections for it. But as noted above, the problem we study is one of model misspecification rather than measurement error. Moreover, almost all strategies in this literature rely on validation data linking ML-generated variables and their ground-truth values.³ In our setting, however, the true dimensions of differentiation are latent, rendering these methods inapplicable.

The remainder of the paper is structured as follows. Section 2 presents our bias corrections and diagnostics for BLP-type models, while Section 3 does the same for models with individual-level choice data and product fixed effects. Both sections

¹[Grieco et al. \(2025\)](#) study efficient estimation of model parameters in mixed logit models with combined market-level and microdata. Our focus is instead on efficient estimation of counterfactuals. For counterfactuals that depend on data moments in addition to model parameters (e.g., average diversions and average welfare), efficient estimators of model parameters do not necessarily lead to efficient estimators of counterfactuals ([Brown and Newey, 1998](#); [Ai and Chen, 2012](#)).

²A recent application of DML in single-product demand estimation is [Bach et al. \(2024\)](#). Unlike our approach, they assume their embeddings perfectly capture the true product attributes and use DML to correct for the first-stage estimation of nuisance functions, not to correct proxy error.

³See, e.g., [Fong and Tyler \(2021\)](#); [Allon et al. \(2023\)](#); [Angelopoulos et al. \(2023\)](#); [Egami et al. \(2023\)](#); [Zhang et al. \(2023\)](#); [Carlson and Dell \(2025\)](#) and references therein. These works build on an earlier literature on auxiliary data ([Chen et al., 2005, 2008](#)). One exception is [Battaglia et al. \(2024\)](#) who develop analytical bias corrections for linear regression without validation data.

conclude with a practitioner’s guide detailing the steps involved and giving practical recommendations. Simulations and the empirical application are presented in Sections 4 and 5, respectively. Section 6 presents all theoretical results with all proofs deferred to Appendix A. Section 7 concludes.

2 Case 1: Endogenous Prices

We first consider a setting where prices vary at the market level and identification is achieved through instruments.

2.1 Model and Data

Following an established literature (Berry and Haile, 2014; Freyberger, 2015), we assume that one has data on a large number T of markets in which (subsets of) J goods are sold.⁴ In addition to the outside option (denoted by 0), each market t features products $\mathcal{J}_t \subseteq \{1, \dots, J\}$, for which prices $p_t = (p_{jt})_{j \in \mathcal{J}_t}$, exogenous product attributes $x_t = (x_{jt})_{j \in \mathcal{J}_t}$, and market shares $s_t = (s_{jt})_{j \in \mathcal{J}_t}$ are observed. Consumer choices are also driven by unobservables $\xi_t = (\xi_{jt})_{j \in \mathcal{J}_t}$. The model predicts market shares as a function of p_t , x_t , ξ_t , and a parameter vector θ :

$$s_{jt} = \sigma_j(p_t, x_t, \xi_t; \theta), \quad j \in \mathcal{J}_t. \quad (1)$$

Prices are endogenous and may be correlated with the unobservables. To address this, we rely on a vector of instrumental variables $z_t = (z_{jt})_{j \in \mathcal{J}_t}$ that satisfy

$$\mathbb{E}[\xi_{jt}|z_{jt}] = 0, \quad j \in \mathcal{J}_t, \quad t = 1, \dots, T. \quad (2)$$

We also accommodate “micro BLP” settings (Berry et al., 2004; Berry and Haile, 2024) where, in addition, individual-level data are available in a subset of markets. The microdata consists of choice indicators $d_{it} = (d_{ijt})_{j \in \mathcal{J}_t}$ taking the value 1 if i chose j in market t and 0 otherwise, and demographics y_{it} that vary at the consumer level, such as income, and/or $\bar{y}_{ijt}$ that vary at the product-consumer level, such as distance between a household’s home and a school or a hospital. We assume the microdata

⁴For simplicity, we assume that J is fixed. It is straightforward to extend our approach to asymptotic thought experiments where J grows slowly with the number of markets T .

are available for a fixed set of markets, labeled $t = 1, \dots, \tau$ without loss.⁵ We treat the microdata as repeated cross sections of size $N_1, \dots, N_\tau$.

This model subsumes many empirical specifications used in the literature. We provide two simple examples below to fix ideas.

Example 1 (BLP). *The utility that individual i derives from good j in market t is*

$$\begin{aligned} u_{ijt} &= \beta'_i x_{jt} - \alpha_i p_{jt} + \xi_{jt} + \varepsilon_{ijt}, & j \in \mathcal{J}_t, \\ u_{i0t} &= \varepsilon_{i0t}, \end{aligned} \quad (3)$$

where the ε_{ijt} are iid type 1 extreme value random variables. Market shares are

$$\sigma_j(p_t, \xi_t, x_t; \theta) = \int \frac{e^{\beta'_i x_{jt} - \alpha_i p_{jt} + \xi_{jt}}}{1 + \sum_{k \in \mathcal{J}_t} e^{\beta'_i x_{kt} - \alpha_i p_{kt} + \xi_{kt}}} dF(\alpha, \beta; \theta), \quad j \in \mathcal{J}_t, \quad (4)$$

for some parametric distribution F .

Example 2 (Micro BLP). *The utility is specified as:*

$$\begin{aligned} u_{ijt} &= \beta'_i x_{jt} - \alpha_i p_{jt} + (x_{jt}, p_{jt})' \Pi y_{it} + \pi' \bar{y}_{ijt} + \xi_{jt} + \varepsilon_{ijt}, & j \in \mathcal{J}_t, \\ u_{i0t} &= \varepsilon_{i0t}, \end{aligned} \quad (5)$$

Micro moments can be computed using the following expression for choice probabilities:

$$\begin{aligned} Pr(d_{ijt} = 1 | y_{it}, \bar{y}_{it}, p_t, \xi_t, x_t; \theta) \\ = \int \frac{e^{\beta'_i x_{jt} - \alpha_i p_{jt} + (x_{jt}, p_{jt})' \Pi y_{it} + \pi' \bar{y}_{ijt} + \xi_{jt}}}{1 + \sum_{k \in \mathcal{J}_t} e^{\beta'_i x_{kt} - \alpha_i p_{kt} + (x_{kt}, p_{kt})' \Pi y_{it} + \pi' \bar{y}_{ikt} + \xi_{kt}}} dF(\alpha, \beta; \theta). \end{aligned} \quad (6)$$

Market shares are obtained by integrating (6) over the (known) distribution of $(y_{it}, \bar{y}_{it})$ for $\bar{y}_{it} = (\bar{y}_{ijt})_{j \in \mathcal{J}_t}$.

So far the model is standard. Our point of departure is to partition

$$x_{jt} \equiv (\bar{x}_{jt}, e_j),$$

where $\bar{x}_{jt}$ is a vector of conventional observed product attributes, such as product size, and e_j is an r -vector of product attributes, such as visual design or user friendli-

⁵The case with microdata for all T markets is simpler. Derivations are available upon request.

ness, that are difficult to capture using standard numeric data. Accordingly, we treat $e = (e'_j)_{j=1}^J$ as known to the consumer but not observed in the data. Instead, the practitioner observes proxies $\tilde{e} = (\tilde{e}'_j)_{j=1}^J$ for the true underlying e . Note that e does not vary across markets, consistent with the fact that difficult-to-quantify characteristics such as visual design or user friendliness are often fixed product characteristics.

Where might $\tilde{e}$ come from? We consider two leading cases:

1. **Unstructured data.** A growing literature computes low-dimensional representations (or *embeddings*) $\tilde{e}_j$ of unstructured data via ML methods. To maximize generality, we stay agnostic on the ML method and the type of unstructured data. It could be text (e.g., product descriptions and reviews), images, audio/video, a combination thereof (Compiani et al., 2025; Han and Lee, 2025), or data from consumer surveys (Magnolfi et al., 2025).
2. **Standard quantitative attributes.** Our approach may equally be used to correct bias when included product attributes that do not vary across markets fail to correctly capture the dimensions of differentiation (e.g., the “mushiness” of cereal hand-coded by Nevo (2001)). In these scenarios, the $\tilde{e}_j$ are proxies for the true latent attributes e_j , and $\bar{x}_{jt}$ represents the remaining attributes that are assumed to be perfectly observed.

Unlike prior work which implicitly treats the $\tilde{e}_j$ as the ground truth, we account for the fact that proxies $\tilde{e}_j$ are only approximations to the true latent e_j . When the $\tilde{e}_j$ are poor proxies for e_j , standard workflows can deliver biased estimates of counterfactuals and invalid inference. Our first main goal is to develop estimators that are immune to this bias. Another goal is to shed light on what a “good” proxy might look like, which can guide the choice of unstructured data source and ML model. This objective is fundamentally different from the standard problem of choosing proxies for a prediction problem, since in our case the counterfactual is not observed in the data.

2.2 Bias Correction for Counterfactuals

We consider a broad class of counterfactuals that can be written as

$$\kappa = \mathbb{E}[k(p_t, \xi_t, \bar{x}_t, e; \theta)], \quad (7)$$

where the expectation is over the distribution of $(p_t, \xi_t, \bar{x}_t)$ across markets t . For instance, κ might represent an average price elasticity, average equilibrium price or consumer welfare measure (possibly after a counterfactual change on the supply side). Counterfactuals for a specific market, such as the price elasticity at a given $(p, \xi, \bar{x})$, are also subsumed. In this case, k is a deterministic function of θ and the expectation becomes redundant. As we discuss below, κ could also represent certain elements of θ , such as the average price coefficient.

In the standard workflow, given the observed aggregate data and a candidate set of proxies $\tilde{e}$, the model parameters θ are first estimated by GMM based on the moment⁶

$$\frac{1}{T} \sum_{t=1}^T Z_t \hat{\xi}_t(s_t, p_t, \bar{x}_t, \tilde{e}; \theta),$$

where $\hat{\xi}_t(s_t, p_t, \bar{x}_t, e; \theta) = (\hat{\xi}_j(s_t, p_t, \bar{x}_t, e; \theta))_{j \in \mathcal{J}_t}$ is defined implicitly via

$$s_{jt} = \sigma_j(p_t, \hat{\xi}_t, \bar{x}_t, e; \theta), \quad j \in \mathcal{J}_t, \quad (8)$$

and Z_t is a $\dim(z) \times |\mathcal{J}_t|$ matrix whose columns contain the original instruments z_{jt} in (2) or transformations thereof, such as optimal instruments (see Remark 4 below). With microdata, the GMM criterion is combined with a minimum-distance criterion based on the micro moments (as in, e.g., [Conlon and Gortmaker, 2025](#)).

Given an estimate $\hat{\theta}$ of θ , the counterfactual κ is usually estimated as

$$\hat{\kappa} = \frac{1}{T} \sum_{t=1}^T k(p_t, \hat{\xi}_t(s_t, p_t, \bar{x}_t, \tilde{e}; \hat{\theta}), \bar{x}_t, \tilde{e}; \hat{\theta}). \quad (9)$$

We call this the *naive estimator* of κ since it does not account for the fact that the proxies $\tilde{e}$ might differ from the true latent attributes e . This *proxy error* has the potential to affect the estimator via two channels: (i) directly, since $\tilde{e}$ is an argument of k , and (ii) indirectly through both $\hat{\theta}$ and $\hat{\xi}_t$.

We now introduce our bias correction procedure. We restrict attention to models in which the attributes e and model parameters θ enter choice probabilities (4) via a

⁶With slight abuse of notation, we now write σ_j as functions of $\bar{x}_t$ and e , with the understanding that σ_j depends on e only through $(e_j)_{j \in \mathcal{J}_t}$, and similarly for $\hat{\xi}_t$.

composite parameter

$$\gamma \equiv \gamma(\theta, e).$$

Many models feature this property. We illustrate it in the BLP example.

Example 1 (continued). We partition $\beta_i = (\beta_{\bar{x},i}, \beta_{e,i})$ and write the utilities as:

$$u_{ijt} = \beta'_{\bar{x},i} \bar{x}_{jt} + \beta'_{e,i} e_j - \alpha_i p_{jt} + \xi_{jt} + \varepsilon_{ijt}, \quad j \in \mathcal{J}_t.$$

Suppose $\alpha_i \sim N(\bar{\alpha}, \sigma_\alpha^2)$, $\beta_{\bar{x},i} \sim N(\bar{\beta}_{\bar{x}}, \Sigma_{\bar{x}})$, $\beta_{e,i} \sim N(\bar{\beta}_e, \Sigma_e)$, and α_i , $\beta_{\bar{x},i}$ and $\beta_{e,i}$ are independent. Then,

$$\theta = (\bar{\alpha}, \sigma_\alpha, \bar{\beta}_{\bar{x}}, \bar{\beta}_e, l(\Sigma_{\bar{x}}), l(\Sigma_e)),$$

where $l(\Sigma)$ stacks the lower-triangular entries of the Cholesky factor of Σ into a vector.⁷ Note that e_j only enters via $\beta'_{e,i} e_j$. Collecting $\beta'_{e,i} e_j$ across products, we have $e\beta_{e,i} \sim N(e\bar{\beta}_e, e\Sigma_e e')$, where $e\Sigma_e e'$ has rank $r \leq J$ because e is $J \times r$. Hence,

$$\gamma(\theta, e) = (\bar{\alpha}, \sigma_\alpha, \bar{\beta}_{\bar{x}}, e\bar{\beta}_e, l(\Sigma_{\bar{x}}), l_r(e\Sigma_e e')),$$

where l_r stacks the lower-triangular entries of the rank- r Cholesky factor of $e\Sigma_e e'$.⁸ For instance, when both $\bar{x}_j$ and e_j are scalars and $J = 2$, we have

$$\gamma(\theta, e) = (\bar{\alpha}, \sigma_\alpha, \bar{\beta}_{\bar{x}}, e_1 \bar{\beta}_e, e_2 \bar{\beta}_e, \sigma_{\bar{x}}, \sigma_e |e_1|, \sigma_e e_2 \text{sign}(e_1)),$$

where $\sigma_{\bar{x}}$ and σ_e are the standard deviations of the random coefficients on $\bar{x}_j$ and e_j .

As can be seen from this example, parameters that do not interact with e , e.g., the average price coefficient $\bar{\alpha}$, are left unchanged in γ . The remaining components of γ capture how e and θ interact to jointly affect utilities. A similar reparameterization for Example 2 is provided in Appendix F.

This reparameterization allows us to simplify notation as follows. First, we note that the right-hand side of (8) depends on (θ, e) only via $\gamma(\theta, e)$. Thus, we write $\hat{\xi}_{jt}(\gamma(\theta, e)) = \hat{\xi}_j(s_t, p_t, \bar{x}_t, e; \theta)$ for $j \in \mathcal{J}_t$ (suppressing dependence on $s_t, p_t, \bar{x}_t$) and let $\hat{\xi}_t(\gamma(\theta, e)) = (\hat{\xi}_{jt}(\gamma(\theta, e)))_{j \in \mathcal{J}_t}$. We similarly restrict attention to counterfactuals

⁷If $\Sigma_{\bar{x}}$ and/or Σ_e are diagonal, then let $l(\Sigma_{\bar{x}})$ and/or $l(\Sigma_e)$ denote their diagonal entries.

⁸As $e\Sigma_e e'$ is $J \times J$ with rank r , $l_r(e\Sigma_e e')$ is the unique $J \times r$ matrix L whose above-diagonal entries are all zeros and whose diagonal entries are all positive, such that $LL' = e\Sigma_e e'$.

that depend on (θ, e) via $\gamma(\theta, e)$ and write $k_t(\gamma(\theta, e)) = k(p_t, \hat{\xi}_t(\gamma(\theta, e)), \bar{x}_t, e; \theta)$. This includes many counterfactuals, such as elasticities with respect to prices or $\bar{x}$, equilibrium prices, and welfare changes associated with changes in prices or $\bar{x}$. It precludes quantifying the effect of changes in e or measuring heterogeneity in preferences for e , but these have little meaning when e has no natural scale or interpretation.

Without microdata, the *bias-corrected estimator* is

$$\hat{\kappa}_{bc} = \frac{1}{T} \sum_{t=1}^T (k_t(\hat{\gamma}) - \hat{c}' Z_t \hat{\xi}_t(\hat{\gamma})), \quad (10)$$

where $\hat{\gamma} = \gamma(\hat{\theta}, \tilde{e})$ denotes the value of γ at the estimated structural parameters $\hat{\theta}$ and the candidate proxies $\tilde{e}$, and $\hat{c}$ is a $\dim(z) \times 1$ vector of weights. We give a closed-form expression for $\hat{c}$ below. This bias correction is easy to implement: it simply takes the naive estimator $\frac{1}{T} \sum_{t=1}^T k_t(\hat{\gamma})$ and adds a weighted average of the estimation moments. Thus, it requires minimal computation beyond what is needed to estimate model parameters θ in the first place.

With microdata, the bias correction also depends on micro moments. Let $\bar{m}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} m_{it}$, where $m_{it} = m(y_{it}, \bar{y}_{it}, d_{it})$ is a known function of demographics and choice data for individual i in market t . Let $m(p_t, \xi_t, \bar{x}_t, e; \theta)$ denote the model-implied expectation of m_{it} conditional on the market-level data:

$$m(p_t, \xi_t, \bar{x}_t, e; \theta) = \mathbb{E}[m_{it} | p_t, \xi_t, \bar{x}_t, e].$$

As the choice probabilities (6) depend on (θ, e) only via $\gamma(\theta, e)$, we write $m_t(\gamma(\theta, e)) = m(p_t, \hat{\xi}_t(\gamma(\theta, e)), \bar{x}_t, e; \theta)$. With this notation, the bias-corrected estimator is

$$\hat{\kappa}_{bc} = \frac{1}{T} \sum_{t=1}^T (k_t(\hat{\gamma}) - \hat{c}' Z_t \hat{\xi}_t(\hat{\gamma})) + \sum_{t=1}^{\tau} \hat{d}_t' (\bar{m}_t - m_t(\hat{\gamma})), \quad (11)$$

where $\hat{c}$ and $\hat{\gamma}$ are as above, and $\hat{d}_1, \dots, \hat{d}_{\tau}$ are $\dim(m) \times 1$ vectors of weights for the micro moments, given below.

The idea behind (10) and (11) is to choose the weights so that $\hat{\kappa}_{bc}$ does not depend on $\hat{\gamma}$ to first order.⁹ This means that, to first order, $\hat{\kappa}_{bc}$ behaves like the right-hand

⁹There is a long tradition of using corrections such as these in two-step estimation. See, e.g., Andrews (1994a) and Newey (1994). Of course, similar debiasing ideas underlie the DML literature.

side of (10) or (11) with $\hat{\gamma}$ replaced by the true value $\gamma_0 = \gamma(\theta_0, e_0)$, where θ_0 are the true structural parameters and e_0 are the true latent attributes. Section 6.1 formally shows that $\hat{\kappa}_{bc}$ is asymptotically unbiased and its distribution does not depend on $\hat{\gamma}$. This has two important implications. First, $\hat{\kappa}_{bc}$ is immune to any bias arising from proxying e with $\tilde{e}$. Second, standard errors do not need to be corrected when $\tilde{e}$ is chosen in a data-dependent way, e.g., by fine tuning an ML model on choice data.

For the intuition, consider the case without microdata. A Taylor expansion of the naive estimator yields

$$\hat{\kappa} \approx \frac{1}{T} \sum_{t=1}^T \left(k_t(\gamma_0) + \frac{\partial k_t(\hat{\gamma})}{\partial \gamma'} (\hat{\gamma} - \gamma_0) \right).$$

We wish to eliminate the second problematic term depending on $\hat{\gamma}$. We replicate this term using the estimation moments, by choosing $\hat{c}$ so that

$$\frac{1}{T} \sum_{t=1}^T \frac{\partial k_t(\hat{\gamma})}{\partial \gamma'} = \hat{c}' \left(\frac{1}{T} \sum_{t=1}^T Z_t \frac{\partial \hat{\xi}_t(\hat{\gamma})}{\partial \gamma'} \right).$$

Substituting in the previous display and “undoing” the Taylor expansion, we get

$$\begin{aligned} \hat{\kappa} &\approx \frac{1}{T} \sum_{t=1}^T \left(k_t(\gamma_0) + \hat{c}' Z_t \frac{\partial \hat{\xi}_t(\hat{\gamma})}{\partial \gamma'} (\hat{\gamma} - \gamma_0) \right) \\ &\approx \frac{1}{T} \sum_{t=1}^T \left(k_t(\gamma_0) + \hat{c}' Z_t \hat{\xi}_t(\hat{\gamma}) - \hat{c}' Z_t \hat{\xi}_t(\gamma_0) \right). \end{aligned}$$

This suggests that to correct bias we want to adjust the naive estimator by subtracting $\frac{1}{T} \sum_{t=1}^T (\hat{c}' Z_t \hat{\xi}_t(\hat{\gamma}) - \hat{c}' Z_t \hat{\xi}_t(\gamma_0))$. The final (infeasible) term depending on γ_0 has mean zero by virtue of (2), so we drop it, leading to the corrected estimator (10). This correction therefore ensures that, to first order, $\hat{\kappa}_{bc}$ depends only on γ_0 .

What assumptions are needed for this result? Besides standard regularity conditions, we require that the discrepancy between $\hat{\gamma}$ and γ_0 not be too large relative to sampling error (see Section 6.1 for a discussion). This is an entirely standard condition in semiparametric two-step estimation (e.g., Newey (1994)) and is unavoidable given the nonlinear nature of multi-product demand models. Further, in simulations calibrated to the Nevo (2001) data, Figure 1 shows that the bias correction is effective even when the proxy error is sufficiently large that the naive estimator is badly biased.

We also implicitly require that the e_j and $\tilde{e}_j$ have the same dimension. Importantly, in the next subsection we provide two diagnostics to help validate both assumptions.

To introduce the expressions for the weights $\hat{c}$ and $\hat{d}_1, \dots, \hat{d}_\tau$, let

$$\begin{aligned}\hat{V} &= \frac{1}{T} \sum_{t=1}^T (Z_t \hat{\xi}_t(\hat{\gamma}))(Z_t \hat{\xi}_t(\hat{\gamma}))' - \bar{g} \bar{g}', \\ \hat{V}_t &= \frac{T}{N_t} \left(\frac{1}{N_t} \sum_{i=1}^{N_t} m_{it} m'_{it} - \bar{m}_t \bar{m}'_t \right), \quad t = 1, \dots, \tau,\end{aligned}$$

denote the sample variance of the estimation moments, where $\bar{g} = \frac{1}{T} \sum_{t=1}^T Z_t \hat{\xi}_t(\hat{\gamma})$. Define

$$\hat{h} = \hat{k} - \hat{G}' \hat{V}^{-1} \hat{K}, \quad \hat{H} = \hat{G}' \hat{V}^{-1} \hat{G} + \sum_{t=1}^{\tau} \hat{M}'_t \hat{V}_t^{-1} \hat{M}_t, \quad (12)$$

where $\hat{k} = \frac{1}{T} \sum_{t=1}^T \dot{k}_t(\hat{\gamma})$ is $\dim(\gamma) \times 1$, $\hat{K} = \frac{1}{T} \sum_{t=1}^T k_t(\hat{\gamma}) Z_t \hat{\xi}_t(\hat{\gamma}) - \hat{\kappa} \bar{g}$ is $\dim(z) \times 1$, $\hat{G} = \frac{1}{T} \sum_{t=1}^T Z_t \dot{\xi}_t(\hat{\gamma})$ is $\dim(z) \times \dim(\gamma)$, $\hat{M}_t = \dot{m}_t(\hat{\gamma})$ is $\dim(m) \times \dim(\gamma)$, and $\dot{k}_t(\gamma) = \frac{\partial k_t(\gamma)}{\partial \gamma}$, $\dot{\xi}_t(\gamma)' = \frac{\partial \xi_t(\gamma)'}{\partial \gamma}$, and $\dot{m}_t(\gamma)' = \frac{\partial m_t(\gamma)'}{\partial \gamma}$ are $\dim(\gamma) \times 1$, $\dim(\gamma) \times J_t$, and $\dim(\gamma) \times \dim(m)$, respectively. The weights to plug into (11) are

$$\hat{c} = \hat{V}^{-1}(\hat{K} + \hat{G} \hat{H}^{-1} \hat{h}), \quad (13)$$

and

$$\hat{d}_t = \hat{V}_t^{-1} \hat{M}_t \hat{H}^{-1} \hat{h}, \quad t = 1, \dots, \tau. \quad (14)$$

Without microdata, $\hat{d}_1 = \dots = \hat{d}_\tau = 0$ and $\hat{H} = \hat{G}' \hat{V}^{-1} \hat{G}$.

We defer formal statements of the results sketched out above to Section 6.1, and instead highlight a few key properties of the corrected estimator.

Remark 1 (Easy to compute standard errors). *The asymptotic variance of the bias-corrected estimator can be estimated as follows:*

$$\hat{V}_{bc} = \hat{s}_k^2 + \hat{c}' \hat{V} \hat{c} - 2 \hat{c}' \hat{K} + \sum_{t=1}^{\tau} \hat{d}'_t \hat{V}_t \hat{d}_t, \quad (15)$$

where $\hat{s}_k^2 = \frac{1}{T} \sum_{t=1}^T k_t(\hat{\gamma})^2 - \hat{\kappa}^2$ is the sample variance of $k_t(\hat{\gamma})$. Without microdata,

the expression simplifies to

$$\hat{V}_{bc} = \hat{s}_k^2 + \hat{c}'\hat{V}\hat{c} - 2\hat{c}'\hat{K}. \quad (16)$$

In either case, standard errors for $\hat{\kappa}_{bc}$ are $\sqrt{\hat{V}_{bc}/T}$. Again, these are closed-form expressions involving quantities that are easy to compute given $\hat{\theta}$.

Remark 2 (Efficiency). While there are many different choice of weights $\hat{c}$ and $\hat{d}_1, \dots, \hat{d}_\tau$ that correct bias, the weights given above are chosen so that the asymptotic variance of $\hat{\kappa}_{bc}$ is as small as possible given the instruments. See Proposition 2 for a formal statement. Importantly, $\hat{\kappa}_{bc}$ remains efficient even if $\hat{\theta}$ is inefficient.

Remark 3 (Fine Tuning). The bias correction and standard error formulas allow the proxies $\tilde{e}$ to be sample-dependent. This accommodates scenarios where an off-the-shelf algorithm has been fine tuned to obtain embeddings that better fit the choice data. Thus, there is no need to correct the standard errors for fine tuning.

Remark 4 (Optimal Instruments). We recommend transforming the instruments z_t in (2) into “optimal instruments” for γ . An advantage of this approach is that the dimension of Z_t is always compatible with that of γ . That is, no additional instruments are needed to implement the bias correction: one just takes different transformations of the original instruments z_t in (2).

To construct optimal instruments for γ , we adapt the approach of Conlon and Gortmaker (2020) with a demand side only. We regress each p_{jt} on the full set of instruments z_t and let $\hat{p}_{jt}$ denote the fitted values. We then compute market shares $\hat{s}_{jt} = \sigma_j(\hat{p}_t, \hat{\xi}_t, \bar{x}_t, \tilde{e}; \hat{\theta})$ at the predicted prices $\hat{p}_t = (\hat{p}_{jt})_{j \in \mathcal{J}_t}$ and $\hat{\xi}_t = 0$. Writing $\tilde{\xi}_{jt}(\gamma(\theta, e)) = \hat{\xi}_j(\hat{s}_t, \hat{p}_t, \bar{x}_t, e; \theta)$ with $\hat{s}_t = (\hat{s}_{jt})_{j \in \mathcal{J}_t}$, we then compute

$$\tilde{z}_{jt} = \frac{\partial \tilde{\xi}_{jt}(\hat{\gamma})}{\partial \gamma}, \quad j \in \mathcal{J}_t.$$

Finally, we concatenate the $\tilde{z}_{jt}$ across products to form the $\dim(\gamma) \times |\mathcal{J}_t|$ matrix Z_t .

Remark 5 (Product or Market Fixed Effects). Our method accommodates additive product and/or market fixed effects in utilities by de-meaning Z_t and $\hat{\xi}_t$ at the product and/or market level before forming moments (see, e.g., Somaini and Wolak (2016) and Conlon and Gortmaker (2020)). All methods can be implemented as described

in this section based on moments formed from the de-meaned variables. The theory developed in Section 6.1 carries over, as discussed further in Appendix B.

2.3 Diagnostics

Next, we propose two diagnostics that practitioners can use to assess the suitability of a candidate set of proxies $\tilde{e}$. The first speaks to whether the $\hat{\gamma} \equiv \gamma(\hat{\theta}, \tilde{e})$ induced by $\tilde{e}$ is sufficiently close to the truth $\gamma_0 \equiv \gamma(\theta_0, e_0)$. The second addresses the question of whether the dimension of $\tilde{e}$ matches that of e_0 . Both diagnostics are based on LM statistics evaluated at $\hat{\gamma}$, so they require minimal additional computation.

2.3.1 Diagnostic 1: Is $\hat{\gamma}$ Close To γ_0 ?

The bias correction is based on linearization, and its theory developed in Proposition 1 below requires the unavoidable assumption that the discrepancy between $\hat{\gamma}$ and γ_0 not be too large relative to sampling error, as discussed above. Here we show that a simple LM statistic can be used to validate this condition. This diagnostic also helps guide the choice of proxies (e.g., embeddings from various data sources or ML methods).

The diagnostic is

$$LM_1 = \|\sqrt{T}\hat{H}^{-1/2}\hat{S}\|^2, \quad (17)$$

where $\hat{S} = \hat{G}'\hat{V}^{-1}(\frac{1}{T}\sum_{t=1}^T Z_t\hat{\xi}_t(\hat{\gamma})) + \sum_{t=1}^T \hat{M}'_t\hat{V}_t^{-1}(m_t(\hat{\gamma}) - \bar{m}_t)$ represents the “score” at $\hat{\gamma}$ and $\hat{H}$ is given in (12). This diagnostic can be interpreted as an LM statistic for testing the null hypothesis that γ_0 is in the set of composite parameters spanned by the candidate proxies $\tilde{e}$, ignoring the fact that $\tilde{e}$ is possibly stochastic.¹⁰

Proposition 4 below shows that LM_1 behaves asymptotically like $\|\sqrt{T}(\hat{\gamma} - \gamma_0)\|^2$. This allows one to validate the assumption that the discrepancy between $\hat{\gamma}$ and γ_0 is sufficiently small, as needed in Proposition 1, by checking whether LM_1 is below a threshold. See the practitioner’s guide at the end of this section for a concrete recommendation, and Section 6.1 for a formal discussion. It can also be shown under a slight strengthening of the conditions of Proposition 4 that LM_1 can be used to bound $\|\tilde{e} - e_0\|^2$, providing a measure of how well the proxies $\tilde{e}$ capture the true latent attributes e driving consumer choices. This is especially useful as the true e_0

¹⁰Formally, a test of $\mathbb{H}_0 : \gamma_0 \in \Gamma(\tilde{e}) := \{\gamma(\theta, \tilde{e}) : \theta \in \Theta\}$ against the alternative $\mathbb{H}_1 : \gamma_0 \in \Gamma \setminus \Gamma(\tilde{e})$, where Θ and Γ are the parameter spaces for θ and γ , respectively.

can never be observed. As a result, LM_1 also serves as a criterion to target when choosing among proxies and/or fine tuning.

2.3.2 Diagnostic 2: Is The Dimension of $\tilde{e}$ Correct?

Practitioners also have to choose the dimension of $\tilde{e}$, i.e., how many proxies to include per product. This is particularly delicate when the proxies don't have a natural economic interpretation, e.g., when they are obtained from black-box algorithms or by applying PCA to many numeric attributes. For example, in the application of Section 5, we use PCA to reduce the dimension of proxies obtained from pre-trained algorithms and need to choose how many principal components to include.

We guide this choice by providing a second diagnostic, denoted LM_2 , again based on an LM statistic. The idea is to augment $\tilde{e}$ with a vector $\eta \in \mathbb{R}^J$ representing some excluded but potentially important product attributes and augment θ with an additional component $\psi \in \Psi$ representing coefficients on η . The diagnostic LM_2 is then an LM statistic for the null that $\psi = 0$. As with LM_1 , it depends on $\hat{\theta}$ only and therefore requires minimal additional computation. Full details are deferred to Appendix C.1.

2.4 Practitioner's Guide

Given a counterfactual of interest κ and a candidate set of proxies $\tilde{e}$:

1. Compute the model parameter estimates $\hat{\theta}$ as usual, treating $\tilde{e}$ as the truth.
2. Compute weights $\hat{c}$ in (13) and, if microdata is available, weights $\hat{d}_t$ in (14).
3. Plug the weights into (11) to compute the bias-corrected estimator $\hat{\kappa}_{bc}$. If only aggregate data is available, use (10) instead.
4. Compute standard errors for $\hat{\kappa}_{bc}$ using the formulas in Remark 1.
5. To check whether a given set of proxies $\tilde{e}$ is adequate:
 - (a) Compute the LM_1 statistic in (17). If it's below a threshold C_T^2 , conclude that the $\hat{\gamma}$ induced by $\tilde{e}$ is sufficiently close to γ_0 . In Section 6.1.2, we motivate a threshold of $C_T^2 = \chi_{\dim(\gamma), 0.95}^2 \log T$.

- (b) Compute the LM_2 statistic in (37). If it's below a critical value $\hat{\xi}_{0.95}^*$, conclude that $\tilde{e}$ is of adequate dimension. The bootstrap method in Appendix C.1 can be used to compute $\hat{\xi}_{0.95}^*$.

2.5 What About Models with Standard Numeric Attributes?

Our approach also provides a way to robustly and efficiently estimate counterfactuals and validate some of the assumptions implicitly made in the demand estimation literature in contexts where only standard numeric attributes are available. The typical workflow implicitly assumes that the product attributes included in the model perfectly capture the dimensions of differentiation. Our bias correction allows practitioners to relax this assumption. To do so, the bias-corrected estimator can be implemented as above, where now $\tilde{e}$ simply represents the numeric attributes that may be imperfect proxies. Similarly, our diagnostics may be used to choose which and how many attributes to include.

Even in the special case that proxy error is not a concern, our bias-corrected estimator and standard error formulas can be used as above with $\gamma \equiv \theta$ to easily perform efficient inference on counterfactuals. This approach offers a few advantages: (i) it yields efficient estimates of counterfactuals (given the instruments) even when $\hat{\theta}$ is inefficient, (ii) standard errors for the counterfactual are available in closed form; and (iii) it allows for combined market-level and microdata.

3 Case 2: Individual-Level Price Variation and Product Fixed Effects

Next, we consider settings with individual-level price variation and choice data. Following an established literature (as reviewed in Dubé and Rossi, 2019), we include product fixed effects to account for systematic differences across products and rule out any remaining price endogeneity.

3.1 Model and Data

We assume that one has data on a large number n of consumers in a single market¹¹ in which J goods are sold,¹² and identify the outside option with $j = 0$. For each consumer i , we observe individual choices $d_i = (d_{ij})_{j=1}^J$ where $d_{ij} = 1$ if i chooses good j and 0 otherwise, $p_i = (p_{ij})_{j=1}^J$ which collects prices and other variables (e.g., rankings on the results page) that vary across consumers, and a vector of demographics y_i . For each product j , the data may also contain attributes x_j that are common across all consumers. The model predicts choice probabilities as a function of $p_i, y_i, x = (x_j)_{j=1}^J$, and a parameter vector θ :

$$Pr(d_{ij} = 1 | p_i, y_i, x; \theta) = \sigma_j(p_i, y_i, x; \theta), \quad j = 1, \dots, J. \quad (18)$$

This setting covers many empirical examples. Here we give just one standard model.

Example 3 (Mixed Logit with Fixed Effects). *The utility that individual i derives from good j is of mixed logit form with microdata:*

$$\begin{aligned} u_{ij} &= \alpha'_i p_{ij} + \beta'_i x_j + x'_j \Pi y_i + \xi_j + \varepsilon_{ij}, \quad j = 1, \dots, J, \\ u_{i0} &= \varepsilon_{i0}, \end{aligned}$$

where ξ_j is a product fixed effect and the ε_{ij} are iid type 1 extreme value random variables. The vector x_j collects characteristics with random coefficients only; characteristics with non-random coefficients are absorbed into the product fixed effect ξ_j . Choice probabilities are

$$\sigma_j(p_i, y_i, x; \theta) = \int \frac{e^{\alpha'_i p_{ij} + \beta'_i x_j + x'_j \Pi y_i + \xi_j}}{1 + \sum_{k=1}^J e^{\alpha'_i p_{ik} + \beta'_i x_k + x'_k \Pi y_i + \xi_k}} dF(\alpha, \beta; \theta), \quad j = 1, \dots, J,$$

where F is a parametric distribution and θ contains $(\xi_j)_{j=1}^J$ and other parameters.

Our point of departure is again to partition $x_j \equiv (\bar{x}_j, e_j)$, where $\bar{x}_j$ collects quantitative product attributes observed by the practitioner, such as product size, and e_j

¹¹For ease of exposition, we present results for a single market, though our approach extends easily to settings with multiple markets.

¹²As before, we assume that J is fixed but it is straightforward to extend our analysis to asymptotic thought experiments where J grows slowly with n .

is an r -vector of product attributes that are not observed in the data. Instead, one has proxies $\tilde{e} = (\tilde{e}'_j)_{j=1}^J$ for the true underlying $e = (e'_j)_{j=1}^J$. We stay agnostic on the form that $\tilde{e}$ takes. A leading case is when $\tilde{e}$ are embeddings computed to represent unstructured data, though our approach may equally be used when $\tilde{e}$ represents any product attributes that fail to correctly capture the true dimensions of differentiation.

3.2 Bias Correction for Counterfactuals

We again consider a broad class of counterfactuals that can be written as

$$\kappa = \mathbb{E}[k(p_i, y_i, \bar{x}, e; \theta)],$$

where the expectation is over the distribution of (p_i, y_i) , and $\bar{x} = (\bar{x}_j)_{j=1}^J$. Section 2.2 lists several counterfactuals of interest that are covered by this class.

In the usual workflow, model parameters are estimated using the observed data and a candidate set of proxies $\tilde{e}$. For instance, one could use maximum likelihood estimation based on

$$\sum_{i=1}^n \sum_{j=0}^J d_{ij} \log \sigma_j(p_i, y_i, \bar{x}, \tilde{e}; \theta),$$

where $\sigma_0 = 1 - \sum_{j=1}^J \sigma_j$ is the probability of the outside option. Given an estimate $\hat{\theta}$ of θ , the counterfactual κ is usually estimated as

$$\hat{\kappa} = \frac{1}{n} \sum_{i=1}^n k(p_i, y_i, \bar{x}, \tilde{e}; \hat{\theta}). \quad (19)$$

As before, we refer to this as the *naive estimator* of κ since it does not account for the fact that the proxies $\tilde{e}$ might differ from the true latent attributes. This proxy error affects the naive estimator directly, as $\tilde{e}$ is an argument of k , and indirectly through the first-stage estimate $\hat{\theta}$, as $\tilde{e}$ enters the likelihood.

We now introduce our bias-correction. Again, we consider models in which e and θ enter choice probabilities (18) via a composite parameter

$$\gamma \equiv \gamma(\theta, e).$$

Many common models have this property. We illustrate it in our leading example.

Example 3 (continued). We partition $\beta_i = (\beta_{\bar{x},i}, \beta_{e,i})$ and $\Pi = [\Pi_{\bar{x}} \Pi_e]$ and write

$$u_{ij} = \alpha'_i p_{ij} + \bar{x}'_j \Pi_{\bar{x}} y_i + e'_j \Pi_e y_i + \xi_j + \beta'_{\bar{x},i} \bar{x}_j + \beta'_{e,i} e_j + \varepsilon_{ij}, \quad j = 1, \dots, J,$$

Suppose $\alpha_i \sim N(\bar{\alpha}, \Sigma_\alpha)$, $\beta_{\bar{x},i} \sim N(0, \Sigma_{\bar{x}})$, and $\beta_{e,i} \sim N(0, \Sigma_e)$, and α_i , $\beta_{\bar{x},i}$, and $\beta_{e,i}$ are independent. Recall the notation l and l_r from Example 1. Then,

$$\theta = (\bar{\alpha}, \xi, v(\Pi_{\bar{x}}), v(\Pi_e), l(\Sigma_\alpha), l(\Sigma_{\bar{x}}), l(\Sigma_e)),$$

where $\xi = (\xi_j)_{j=1}^J$, and $v(\Pi)$ stacks the entries of Π into a vector. Collecting $\beta'_{e,i} e_j$ across products, we have $e \beta_{e,i} \sim N(0, e \Sigma_e e')$, where $e \Sigma_e e'$ has rank $r \leq J$. Hence,

$$\gamma(\theta, e) = (\bar{\alpha}, \xi, v(\Pi_{\bar{x}}), v(e \Pi_e), l(\Sigma_\alpha), l(\Sigma_{\bar{x}}), l_r(e \Sigma_e e')).$$

As before, parameters that do not interact with e , such as $\bar{\alpha}$ and ξ , are left unchanged in γ . The remaining components of γ capture how e and θ interact to jointly affect utilities.

We assume in what follows that choice probabilities σ and the counterfactual function k depend on (θ, e) via $\gamma(\theta, e)$, and write

$$\begin{aligned} \sigma_{ij}(\gamma(\theta, e)) &= \sigma_j(p_i, y_i, \bar{x}, e; \theta), \quad j = 1, \dots, J, \\ k_i(\gamma(\theta, e)) &= k(p_i, y_i, \bar{x}, e; \theta). \end{aligned}$$

We then define the *bias-corrected estimator*

$$\hat{\kappa}_{bc} = \frac{1}{n} \sum_{i=1}^n (k_i(\hat{\gamma}) + \hat{c}'_i (d_i - \sigma_i(\hat{\gamma}))), \quad (20)$$

where $\hat{\gamma} = \gamma(\hat{\theta}, \tilde{e})$ is the value of γ pinned down by $\tilde{e}$ and the first-stage estimate $\hat{\theta}$, and $\hat{c}_i$ is a J -vector of weights given by

$$\hat{c}_i = V_i(\hat{\gamma})^{-1} \dot{\sigma}_i(\hat{\gamma}) \hat{H}^{-1} \hat{k}, \quad (21)$$

$d_i = (d_{ij})_{j=1}^J$ and $\sigma_i(\gamma) = (\sigma_{ij}(\gamma))_{j=1}^J$ are J -vectors, $V_i(\gamma) = \text{diag}(\sigma_i(\gamma)) - \sigma_i(\gamma) \sigma_i(\gamma)'$ is $J \times J$, $\hat{H} = \frac{1}{n} \sum_{i=1}^n \dot{\sigma}_i(\hat{\gamma})' V_i(\hat{\gamma})^{-1} \dot{\sigma}_i(\hat{\gamma})$ is $\dim(\gamma) \times \dim(\gamma)$, $\hat{k} = \frac{1}{n} \sum_{i=1}^n \dot{k}_i(\hat{\gamma})$ and $\dot{k}_i(\gamma) = \frac{\partial k_i(\gamma)}{\partial \gamma}$ are $\dim(\gamma) \times 1$, and $\dot{\sigma}_i(\gamma)' = \frac{\partial \sigma_i(\gamma)'}{\partial \gamma}$ is $\dim(\gamma) \times J$. Importantly, $\hat{\kappa}_{bc}$

requires minimal additional computation, as it involves closed-form expressions of objects that can be easily calculated given $\hat{\theta}$.

As before, $\hat{\kappa}_{bc}$ takes the naive estimator and adds an adjustment term that purges the effect of $\hat{\gamma}$ on $\hat{\kappa}_{bc}$ to first order. Proposition 5 below shows that $\hat{\kappa}_{bc}$ is asymptotically unbiased and its distribution does not depend on $\hat{\gamma}$. We highlight a few key implications of this result.

Remark 6 (Easy to compute standard errors). *The asymptotic variance of $\hat{\kappa}_{bc}$ can be easily estimated using*

$$\hat{V}_{bc} = \hat{s}_k^2 + \frac{1}{n} \sum_{i=1}^n \tilde{c}_i' V_i(\hat{\gamma}) \tilde{c}_i, \quad (22)$$

where $\hat{s}_k^2$ is the sample variance of $k_i(\hat{\gamma})$. Standard errors for $\hat{\kappa}_{bc}$ are then $\sqrt{\hat{V}_{bc}/n}$.

Remark 7 (Fine Tuning). *The bias correction in (20) allows the proxies $\tilde{e}$ to be sample-dependent, for instance because an ML algorithm has been fine tuned on the choice data to provide a better fit. As before, there is no need to correct the standard errors: one can use $\sqrt{\hat{V}_{bc}/n}$ with $\hat{V}_{bc}$ as above whether or not $\tilde{e}$ is data-dependent.*

Remark 8 (Semiparametric Efficiency). *We may view model (18) as a conditional moment model based on the moment condition*

$$\sigma_i(\gamma_0) = \mathbb{E}[d_i | p_i, y_i, \bar{x}]. \quad (23)$$

In Section 6.2, we show that the asymptotic variance of $\hat{\kappa}_{bc}$ is the semiparametric efficiency bound for estimating κ in model (23) (Brown and Newey, 1998; Ai and Chen, 2012). Thus, there do not exist regular estimators of κ in model (23) with smaller asymptotic variance than the bias-corrected estimator $\hat{\kappa}_{bc}$.

3.3 Diagnostics

Here we provide variants of the two diagnostics introduced in Section 2.3 that can be used to assess the suitability of a candidate set of proxies $\tilde{e}$.

3.3.1 Diagnostic 1: Is $\hat{\gamma}$ Close To γ_0 ?

As before, a simple LM statistic can be used to validate whether $\hat{\gamma}$ is sufficiently close to γ_0 , as required by the theory for $\hat{\kappa}_{bc}$ in Proposition 5. This diagnostic also

helps guide the choice of proxies (e.g., embeddings from various data sources or ML models). The first diagnostic is

$$LM_1 = \|\sqrt{n}\hat{H}^{-1/2}\hat{S}\|^2, \quad (24)$$

where $\hat{S} = \frac{1}{n} \sum_{i=1}^n \sum_{j=0}^J \frac{d_{ij}}{\sigma_{ij}(\hat{\gamma})} \dot{\sigma}_{ij}(\hat{\gamma})$ is the score and $\hat{H}$ is the (expected) Hessian, defined below (21). Proposition 6 below shows that LM_1 behaves like $\|\sqrt{n}(\hat{\gamma} - \gamma_0)\|^2$ as n grows large. This allows researchers to validate the assumption that the discrepancy between $\hat{\gamma}$ and γ_0 is sufficiently small, as needed in Proposition 5, by checking whether LM_1 is below a threshold. See the practitioner’s guide below for a concrete recommendation and Section 6.2 for a formal discussion. Further, Proposition 7 shows that LM_1 can be used to bound the discrepancy between $\tilde{e}$ and e_0 . Thus, LM_1 also serves as a criterion to target when choosing among proxies and/or fine tuning.

3.3.2 Diagnostic 2: Is The Dimension of $\tilde{e}$ Correct?

We again provide guidance on how many proxies to include per product with a second diagnostic, denoted LM_2 . The diagnostic is based on an LM statistic for testing whether $\tilde{e}$ is of sufficient dimension, and its construction follows similar ideas to Section 2.3.2. As with LM_1 , it depends only on $\hat{\theta}$ and therefore requires minimal additional computation. Full details are deferred to Appendix C.2.

3.4 Practitioner’s Guide

Given a counterfactual of interest κ and a candidate set of proxies $\tilde{e}$:

1. Compute the model parameter estimates $\hat{\theta}$ as usual, treating $\tilde{e}$ as the truth.
2. Compute weights $\hat{c}$ in (21).
3. Plug the weights into (20) to compute the bias-corrected estimator $\hat{\kappa}_{bc}$.
4. Compute standard errors for $\hat{\kappa}_{bc}$ using the formula in Remark 6.
5. To check whether a given set of proxies $\tilde{e}$ is adequate:
 - (a) Compute the LM_1 statistic in (24). If it’s below a threshold C_n^2 , conclude that the $\hat{\gamma}$ induced by $\tilde{e}$ is sufficiently close to γ_0 . In Section 6.2.2, we motivate a threshold of $C_n^2 = \chi_{\dim(\gamma), 0.95}^2 \log n$.

- (b) Compute the LM_2 statistic in (39). If it's below a critical value $\hat{\xi}_{0.95}^*$, conclude that $\tilde{e}$ is of adequate dimension. The bootstrap method in Appendix C.2 can be used to compute $\hat{\xi}_{0.95}^*$.

4 Simulations

We first illustrate our approach in simulations, considering each of the two models discussed in Sections 2 and 3 in turn.

4.1 Case 1: Endogenous Prices

We start by considering the BLP-type model of Section 2. As proof of concept, we use a DGP calibrated to the standard Nevo (2001) ready-to-eat cereal market data, based on the replication data in the PyBLP tutorial (Conlon and Gortmaker, 2020) (see also Nevo (2000b)). For each market, the data contain the market shares of $J = 24$ products, their prices, their sugar content and a dummy variable capturing whether a product is “mushy” or not. First, we fit a mixed logit model to the data using PyBLP.¹³ For 1,000 simulations, we draw prices and exogenous variables for $T = 1,200$ markets¹⁴ from a distribution calibrated to match the mean and variance in the data, and generate market shares using parameters similar to those estimated in the first step (full details of the DGP are in Appendix D.1).¹⁵ We consider the possibility that the mushy variable is an imperfect proxy, which seems plausible given that the variable was hand-coded by the author based on his own judgment. For each simulated data set and each product, we generate data treating the binary mushy variable in the original data as the truth (e_j), and estimate model parameters and counterfactuals based on a corrupted version given by $\tilde{e}_j \sim \text{Beta}(\rho, 2 - \rho)$ if $e_j = 0$, and $\tilde{e}_j \sim \text{Beta}(2 - \rho, \rho)$ if $e_j = 1$ for $\rho = 0, 0.1, 0.2, \dots, 0.6$. The case $\rho = 0$ corresponds to $e_j = \tilde{e}_j$ (no proxy error) while larger values of ρ correspond to more severe error.¹⁶

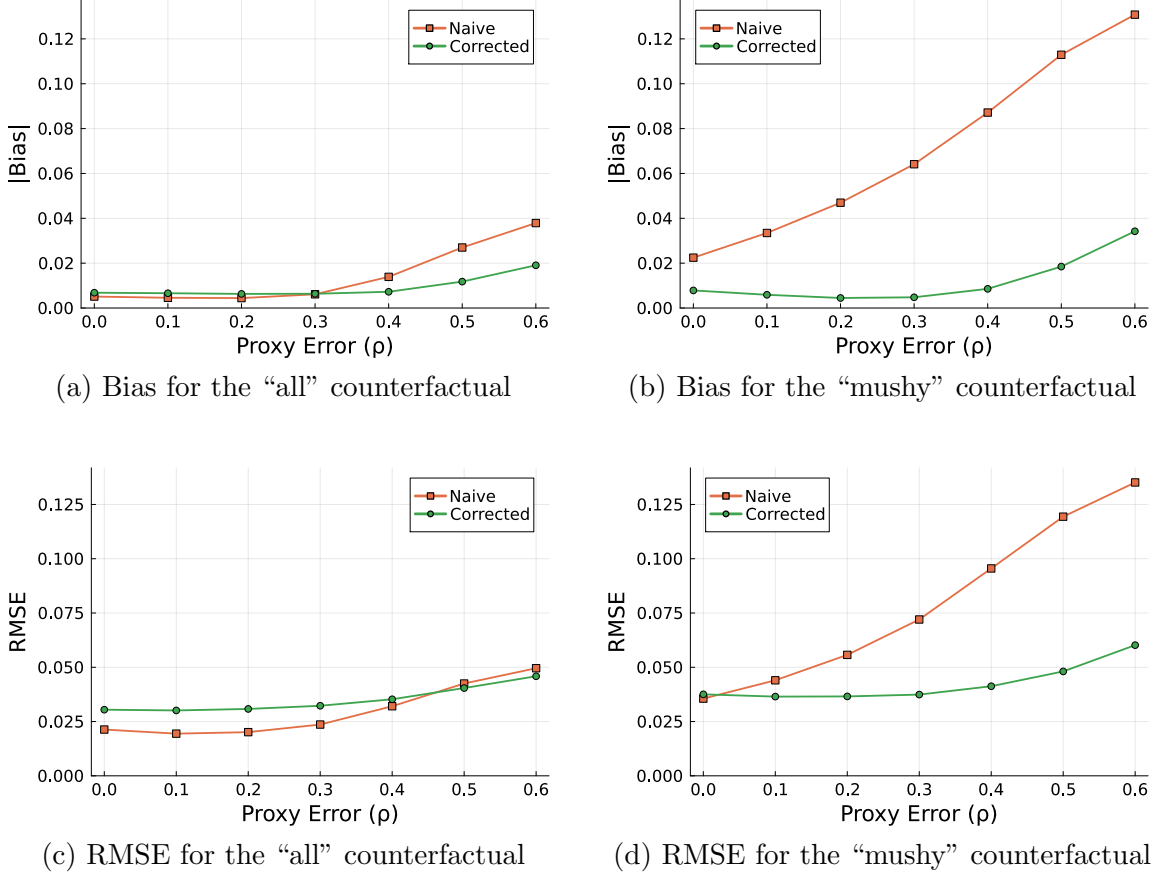
¹³We use the third model estimated in the PyBLP tutorial, which has random coefficients on price, sugar content, the mushy dummy and the constant for the inside products, with a diagonal covariance matrix. Product fixed effects are included.

¹⁴We choose $T = 1,200$ since that is roughly the sample size ($T = 1,124$) in Nevo (2001).

¹⁵As shown in the PyBLP tutorial, the instruments provided with the dataset are not enough to identify the standard deviations on the random coefficients well. To address this, we form additional differentiation IVs following the approach of Gandhi and Houde (2019); see Appendix D.

¹⁶The corrupted embeddings $\tilde{e}_j$ have mean $\rho/2$ and $1 - \rho/2$ if $e_j = 0$ and $e_j = 1$, respectively. Their distributions approach a uniform distribution on $[0, 1]$ as $\rho \rightarrow 1$.

Figure 1: Bias and RMSE as a function of proxy error, [Nevo \(2001\)](#) design



This simulation design matches the scale of one of the canonical applications of the BLP model, allowing us to illustrate the performance of our approach on standard datasets.

We focus on the effects of proxy error on two counterfactuals: (i) the diversion from *all* products of firm 1 to *all* products of firm 2; (ii) the diversion from all *mushy* products of firm 1 to all *mushy* products of firm 2. Diversions are a key metric to understand the nature of competition in a market ([Conlon and Mortimer, 2021](#)).

The left two panels of Figure 1 show the absolute bias and RMSE of the naive estimator $\hat{\kappa}$ in (9) and the bias-corrected estimator $\hat{\kappa}_{bc}$ in (10) as a function of the level of proxy error ρ for the “all” counterfactual. As firms 1 and 2 sell both mushy and non-mushy cereal, this counterfactual involves aggregating across the two types of products, resulting in small bias except for large values of ρ , where the correction reduces bias by around 50%. For smaller values of ρ the correction marginally increases

variance, as evidenced by the slight increases in RMSE relative to the naive estimator. However, for larger values of ρ this increase is more than offset by the bias reduction, and the bias-corrected estimator has lower RMSE than the naive estimator.

The right two panels of Figure 1 show the results for the “mushy” counterfactual. Here, the naive estimator has much larger bias than the bias-corrected estimator across all levels of proxy error. Indeed, the bias-corrected estimator is essentially unbiased for $\rho = 0, 0.1, \dots, 0.4$, while for larger ρ the bias is only a fraction of that of the naive estimator. To see this in more detail, histograms of the naive and bias-corrected estimators for $\rho = 0.1$ and $\rho = 0.3$ are shown in Figure 2. The distribution of the bias-corrected estimator is stable and centered around the truth, while the distribution of the naive estimator drifts to the left for larger ρ . The RMSE of the corrected estimator is uniformly smaller, except in the knife-edge case of no proxy error where it’s essentially on par with the naive estimator. Finally, Table 1 displays the coverage of 95% confidence intervals for the counterfactuals based on the bias-corrected estimator and standard errors computed using the formula (16). Coverage is very close to nominal coverage for $\rho = 0, 0.1, \dots, 0.4$, while for larger values of ρ the bias eventually distorts coverage. Nevertheless, these results show our formulas deliver easy-to-implement and valid inference on counterfactuals, and inference is robust to degrees of proxy error that can severely distort naive estimators.

Interestingly, we obtain very similar results when we include additional differentiation IVs based on the distance between products in the space of (possibly mismeasured) $\tilde{e}_j$. See Appendix D.2 for additional details and results. While our theory does not explicitly cover this case, this finding suggests that researchers might be able to continue using standard BLP-type instruments even when the non-price attributes are imperfect proxies.

4.2 Case 2: Individual-Level Price Variation and Product Fixed Effects

Next, we consider the model in Section 3 with 10 products and 10,000 consumers. These figures are in line with the data used in the empirical application in Section 5. We model utility as a function of price and two-dimensional latent attributes e (in addition to idiosyncratic shocks). The latent attributes e_j are drawn iid $N(0, 1)$ across products. Individual-level prices are drawn iid from a $N(5, 1)$ distribution and vary

Figure 2: Distribution of the naive and bias-corrected estimators, [Nevo \(2001\)](#) design

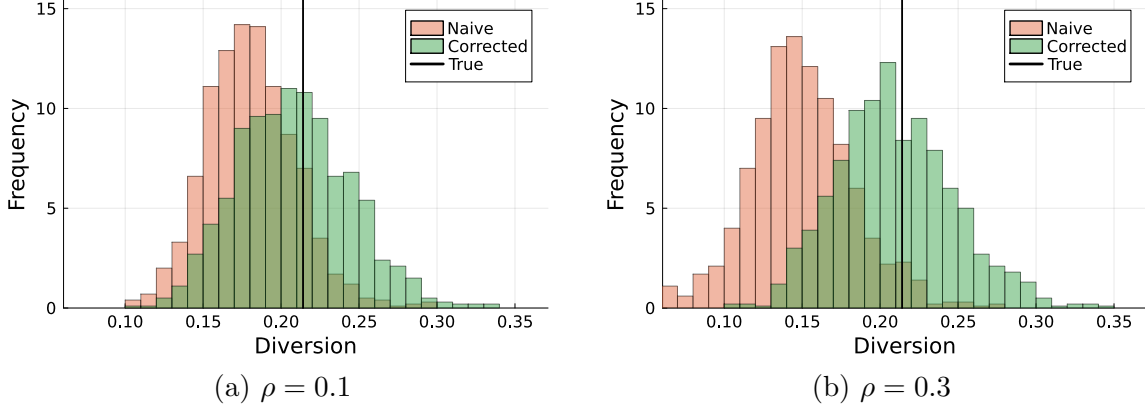


Table 1: Coverage of 95% confidence intervals for counterfactuals, [Nevo \(2001\)](#) design

ρ	Coverage: “all”	Coverage: “mushy”
0.00	0.956	0.967
0.10	0.963	0.972
0.20	0.967	0.977
0.30	0.962	0.971
0.40	0.954	0.943
0.50	0.924	0.880
0.60	0.897	0.773

across simulations. The random coefficient on price is $N(-1, 0.3^2)$ and the coefficients on the latent attributes are iid $N(0, 0.75^2)$. We keep the latent attributes e fixed in the data generating process and vary the amount of error in the proxies $\tilde{e}$ that are used in estimation of θ and counterfactuals. Specifically, for every j , we let:

$$\tilde{e}_j = (1 - \rho/2)e_j + \sqrt{1 - (1 - \rho/2)^2}\eta_j, \quad (25)$$

where η_j is a two-dimensional standard normal random vector drawn iid across simulations and ρ determines the amount of error in $\tilde{e}$. When $\rho = 0$, the proxies exactly match the latent attributes, whereas as ρ increases towards 1 the proxies are increasingly mismeasured. We note that this simulation design imposes very few restrictions on the form of proxy error. In particular, depending on the draw of η_j , each element of $\tilde{e}_j$ could be smaller or larger than the corresponding element of e_j and this can freely

vary across goods j .¹⁷ Finally, we include product fixed effects in estimation but set these to zero when generating data. This ensures that the only source of product-level differentiation is through the embeddings and makes the bias correction task more challenging.

We focus on estimation of diversion, i.e., the fraction of consumers that switch from one product to another one when the former is removed from the choice set. We focus on diversion from product 3 to 4 as this pair has the smallest diversion at the true parameters and embeddings, so correcting bias is (potentially) most critical.

Figure 3 plots histograms of the naive estimator (19) our bias-corrected estimator (20) across 1,000 simulations. As the level of proxy error increases, the distribution of the naive estimator moves away from the true value of the counterfactual (roughly 0.05), whereas the corrected estimator remains centered around the true value. Interestingly, for larger levels of proxy error, the distribution of the naive estimator ends up being centered around the counterfactual prediction of the logit model with no random coefficients. This is intuitive: as the proxies $\tilde{e}$ become increasingly noisy, they capture less of the substitution patterns in the data, and the estimated variance of their random coefficients shrinks towards zero, i.e. the model collapses to one without random coefficients. This finding also serves as a warning that proxy error can defeat the purpose of estimating a random coefficients model in the first place: if the proxy error bias is not properly accounted for, the model may revert to the restrictive substitution patterns that the model was specifically intended to relax.

To better assess the trade-offs involved in our bias correction, Figure 4 shows how the bias and RMSE of the two estimators vary with ρ . When $\tilde{e}$ is a perfect proxy, both the naive and bias-corrected estimators have very low bias. Our estimator has a marginally higher RMSE, indicating that its variance is slightly higher than that of the naive estimator. This is intuitive: since the naive estimator leverages the assumption that the proxies $\tilde{e}$ are correct and ours does not, we obtain slightly less precise estimates when that assumption happens to be correct. However, this is a knife-edge case. When proxy error is present, our estimator consistently achieves lower bias and RMSE than the naive estimator. As expected, when the proxy error becomes very large ($\rho > 0.5$), the bias corrected estimator starts to also incur small

¹⁷Note that (25) is such that $\tilde{e}_j$ has roughly the same amount of variation across goods j as e_j does. This allows us to isolate the effect of proxy error in a way that is not confounded by changes in the scale of the proxies $\tilde{e}_j$ used in estimation.

Figure 3: Distribution of the naive and bias-corrected estimators, [CMS](#) design

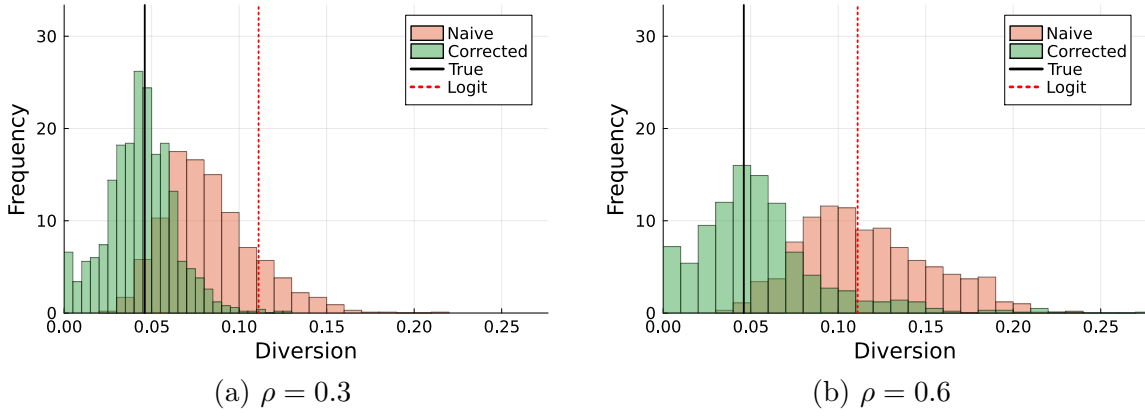


Table 2: Coverage of 95% confidence intervals for counterfactuals, [CMS](#) design

ρ	Coverage
0.00	0.957
0.10	0.957
0.20	0.929
0.30	0.910
0.40	0.893
0.50	0.863
0.60	0.816

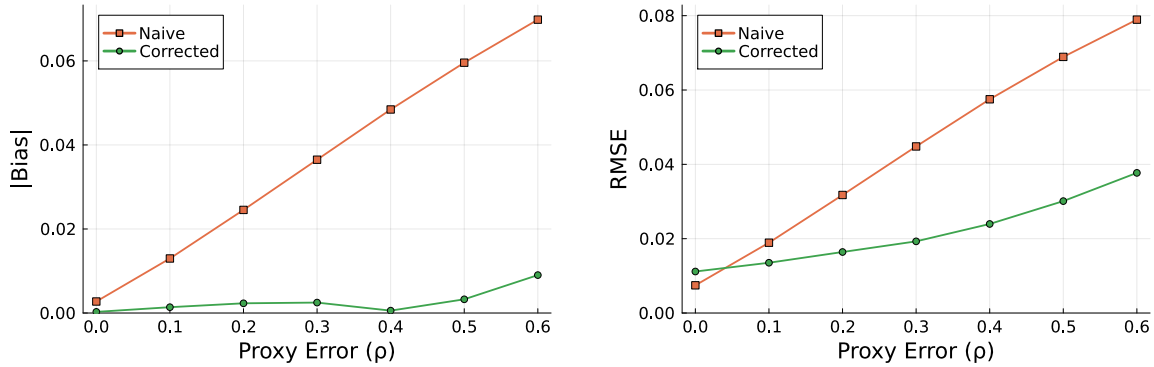
bias.

Finally, Table 2 presents coverage of 95% confidence intervals formed from our bias-corrected estimator (20) with standard errors computed using the formula (22). These have coverage close to nominal coverage for $\rho = 0, 0.1, \dots, 0.3$, while coverage inevitably breaks down when the proxies become very noisy. As with the previous set of simulations, however, these results show that our formulas are again able to deliver valid inference with noisy proxies that otherwise badly bias naive estimators.

5 Empirical Application

We now apply our method to the experimental data from [CMS](#). The data set records the choices made by 9,265 participants when faced with a choice of ten e-books. In a first task, participants were asked to choose their preferred e-book based on informa-

Figure 4: Bias and RMSE of the naive and bias-corrected estimators, CMS design



tion displayed to them, including (randomized) prices, standard attributes (author, year of publication, genre and number of pages), and unstructured information (cover images, titles, plot descriptions and reviews). In a second task, each participant’s first choice was removed and they were asked to choose again from the remaining nine books. CMS estimate a range of models on the first choice data and compare their performance in predicting second choices. This gives a direct measure of how well different models capture counterfactual substitution patterns. Specifically, the paper compares mixed logit models based on standard attributes with mixed logit models that leverage proxies extracted from unstructured data. The key findings are that (i) unstructured data is predictive of substitution patterns, and (ii) book reviews, when processed with transformer-based text models, perform particularly well at predicting substitution.

The results in CMS treat the proxies as if they were correctly specified. However, there are good reasons to believe that proxy error might play an important role. First, the unstructured data are processed using pre-trained ML models that are not targeted towards predicting substitution patterns.¹⁸ While the resulting proxies are found to be predictive of substitution patterns, they may not perfectly capture the underlying attributes that drive consumers’ choices. Second, the dimension of the proxies is reduced via PCA before inputting them into the demand model, which is likely to introduce further proxy error. This also raises the question of how many principal components should be included in the model.

¹⁸Specifically, images are processed via classification models trained to assign each image to one of many classes; texts are processed using bag-of-words and transformer models: the former simply capture word frequency, whereas the latter are trained to predict the next word in a text.

Here we investigate whether applying our bias correction method and diagnostics helps better capture substitution patterns. We use the approach from Section 3 since we have individual-level data and prices are randomized, so endogeneity is not a concern. We focus on the ability of different models to correctly predict the closest substitute for any given book. The second choice data give us a direct measure of this: for a given book A , its closest substitute is the book that most people switch to when A is removed from the choice set.

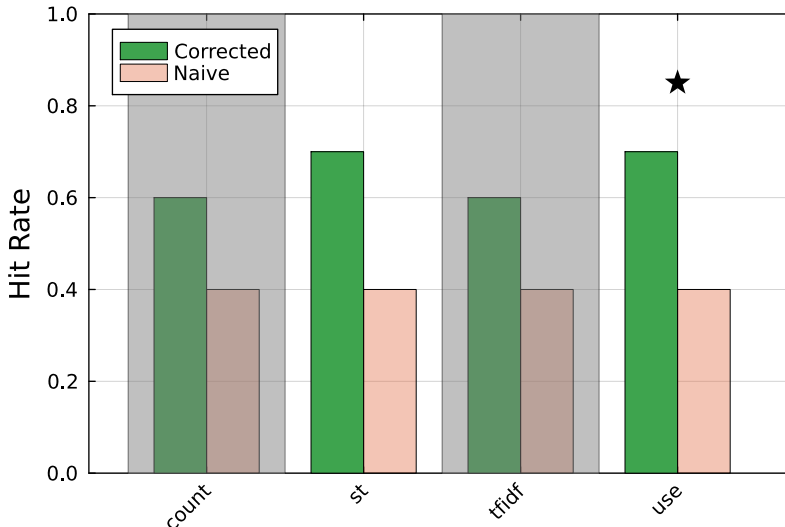
We note that this exercise sets a high bar for our approach. Unlike in a simulation, the demand model might be misspecified even if the proxies are correctly specified. For instance, the model assumes a normal distribution for the random coefficients but the true distribution of preference heterogeneity might be different. As a result, this exercise tests whether our approach works well even in cases where all assumptions needed for the theoretical results might not hold exactly. By looking at substitution patterns in response to product removals that are not part of the estimation data, this provides a direct test of the model’s ability to correctly predict counterfactuals.

Figure 5 shows the results for the proxies extracted from book reviews using four ML models.¹⁹ For each model, we report the fraction of the ten books for which the model correctly identifies the closest substitute (as measured by the second choice data).²⁰ The lighter orange bars show the performance of the naive approach that uses the estimates from *CMS*, whereas the darker green bars show the performance of the bias-corrected estimator. The LM_2 diagnostic rules out two sets of proxies, indicating that they don’t feature a sufficient number of random coefficients. Interestingly, these proxies are extracted using very coarse language models: *Count* is a bag-of-words model that represents text as fixed-length vectors based on simple word counts, whereas *TF-IDF* (Term Frequency-Inverse Document Frequency) is a variation of the bag-of-words approach that assigns more weight to words that are frequent in a specific text but rare in general. By contrast, the proxies that survive the LM_2 diagnostic (*ST* for BERT Sentence Transformer, and *USE* for Universal Sentence Encoder) are extracted using more sophisticated models based on the transformer

¹⁹We focus on book reviews since *CMS* show that reviews are the most predictive of substitution. Appendix E presents the results for the proxies extracted from the other data sources.

²⁰The model prediction of the closest substitute is the product with the highest average (across consumers) second-choice probability once product A is removed. Here, κ corresponds to the probability that B is a consumer’s second choice conditional on A being their first choice, averaged across consumers, which we compute across all (A, B) pairs.

Figure 5: Rates of correct closest-substitutes predictions.



Notes: Darker green bars show the fraction of books for which the bias-corrected estimator correctly identifies the closest substitute. Lighter orange bars show the corresponding fractions for the naive estimator. Each pair of bars corresponds to an ML model used to extract the proxies from the book reviews. The specifications ruled out by the LM_2 diagnostic are grayed out. A star flags the specification with the smallest LM_1 diagnostic.

architecture that underpins modern large language models. The LM_1 diagnostic selects *USE* as the best-fitting model. Further, for both *ST* and *USE*, the fraction of correctly predicted substitutes goes from 40% without the bias correction to 70% with the bias correction, a substantial improvement in performance (for comparison, a coin flip would achieve a hit rate of 11%).

These results confirm that our approach is able to guide researchers towards the most informative proxies and meaningfully improve counterfactual predictions.

5.1 Why Does the Correction Help?

To shed more light on the mechanism underlying the improved performance of the corrected estimator, we take the preferred *USE* proxies and focus on the three prediction tasks in which the bias correction improves over the naive estimator. For each of the three cases, Figure 6 shows what the naive and corrected estimators predict is the closest substitute to the focal book, along with what is revealed empirically by the second-choice data.

The naive model incorrectly predicts the same book, *Please Tell Me*, as the closest

Figure 6: Closest substitute predictions for the three books where the naive and corrected models disagree.

Focal Book	 The Housemaid	 The Serpent & The Wings of Night	 The Ashes & The Star Cursed King
Empirical	 The Inmate	 Court of Ravens and Ruin	 The Serpent & The Wings of Night
Naive (Review USE)	 Please Tell Me	 Please Tell Me	 Please Tell Me
Corrected (Review USE)	 The Inmate	 Court of Ravens and Ruin	 The Serpent & The Wings of Night

Notes: The three columns correspond to the focal books for which the naive and corrected estimators (based on the *Review USE* specification) make different closest-substitute predictions. The row labeled “Empirical” shows the closest substitute to the focal book revealed by the second-choice data.

substitute to all three focal books. Importantly, *Please Tell Me* is the most popular book in the first-choice data. This is a classic Independence of Irrelevant Alternatives (IIA) pattern: the model predicts substitution to the remaining books in proportion to their market shares, regardless of how similar they are to the focal book (McFadden, 1974). Why does this happen? Intuitively, when the proxies (in this case the reviews embeddings) are noisy measures of the underlying dimensions of differentiation driving consumer choices, the mixed logit model will not find the proxies very helpful in capturing substitution and thus will revert toward the IIA-like substitution patterns of a logit model without random coefficients. By contrast, the corrected model breaks this pattern: it redirects substitution toward more similar books and correctly identifies the closest substitute.

This illustration offers what we think is a valuable message for practitioners: using imperfect proxies can defeat the very purpose of estimating a mixed logit model by pushing its predictions towards IIA. Correcting the bias induced by proxy error alleviates this and, at least in our application, leads to a substantial improvement in the model’s counterfactual predictions.

6 Theory

6.1 Case 1: Endogenous Prices

Let Γ denote the set of all values of $\gamma(\theta, e)$ as θ varies over the parameter space Θ and e varies over all $J \times r$ matrices whose first r rows are linearly independent. We shall implicitly assume in what follows that the true latent attributes e_0 and the proxies $\tilde{e}$ are $J \times r$ matrices whose first r rows are linearly independent.²¹ We shall also implicitly assume that Γ is convex and open. This is true for the $\gamma(\theta, e)$ given in Examples 1 and 2, for which Γ is the product of copies of $\mathbb{R}$ and $(0, \infty)$ and is therefore convex and open.²² In Section 2, the counterfactual function k_t , $\hat{\xi}_t$ from (8), and micro moments m_t depended on (θ, e) only through the value of $\gamma(\theta, e)$. We can therefore view k_t , $\hat{\xi}_t$, and m_t as random functions defined on Γ .

6.1.1 Theory for Bias Correction

We first give results for the case of combined market-level data and microdata. We let $\chi_t = (p_t, \xi_t, \bar{x}_t, e)$ and let $\mathcal{M}$ denote the σ -algebra generated by $\chi_1, \dots, \chi_\tau$. Let $V = \text{Var}(Z_t \hat{\xi}_t(\gamma_0))$ be the $\dim(z) \times \dim(z)$ covariance matrix of the aggregate moments at the true parameters, and let $V_t = \text{Var}(m_{it} | \chi_t)$ be the $\dim(m) \times \dim(m)$ covariance matrix of the micro moments for market t conditional on χ_t . Let

$$H = G'V^{-1}G + \sum_{t=1}^{\tau} M_t'(r_t V_t)^{-1} M_t, \quad (26)$$

²¹This is without loss of generality: we can always relabel the products so that the first rows of e_0 are linearly independent provided e_0 has full column rank r .

²² For instance, the operation l stacking the lower-triangular entries of the Cholesky factor maps the manifold of symmetric positive definite matrices into the product of copies of $\mathbb{R}$ and $(0, \infty)$. A similar result holds for the reduced-rank Cholesky decomposition l_r (Neuman et al., 2023).

where $r_1, \dots, r_\tau$ are defined in Assumption 1 below, and let $h = \mathbb{E}[\dot{k}_t(\gamma_0)] - G'V^{-1}K$, where $K = \mathbb{E}[k_t(\gamma_0)Z_t\hat{\xi}_t(\gamma_0)]$ is $\dim(z) \times 1$, $G = \mathbb{E}[Z_t\dot{\xi}_t(\gamma_0)]$ is $\dim(z) \times \dim(\gamma)$, and $M_t = \dot{m}_t(\gamma_0)$ is $\dim(m) \times \dim(\gamma)$. Here V and h are deterministic whereas $V_1, \dots, V_\tau$ and H are $\mathcal{M}$ -measurable random matrices. Let N be a neighborhood of γ_0 .

Assumption 1. Let the following hold:

- (i) $k_t(\cdot)$, $m_t(\cdot)$, and $Z_t\hat{\xi}_t(\cdot)$ are twice continuously differentiable in γ on N (almost surely), and elements of the functions and their first and second derivatives are uniformly (for $\gamma \in N$) bounded by a random variable D_t with finite fourth moment;
- (ii) $\mathbb{E}[\|m_{it}\|^{2+\delta}|\mathcal{M}] \leq C$ (almost surely) for some $0 < \delta, C < \infty$;
- (iii) V is positive definite and $\lambda_{\min}(V_1), \dots, \lambda_{\min}(V_\tau), \lambda_{\min}(H) \geq \epsilon$ (almost surely) for some $\epsilon > 0$;
- (iv) $T/N_t \rightarrow r_t \in (0, \infty)$ for $1 \leq t \leq \tau$.

Parts (i)-(iii) of Assumption 1 are standard smoothness, moment, and rank conditions, respectively. Assumption 1(iv) requires that the aggregate data and microdata have comparable sample sizes. This is designed to accommodate common empirical scenarios where T is in the high tens or hundreds and N_t is in the hundreds or thousands for each market (e.g., Petrin (2002) and Grieco et al. (2024)).

The next result shows that the bias-corrected estimator $\hat{\kappa}_{bc}$ from (11) is asymptotically centered at the true counterfactual $\kappa_0 = \mathbb{E}[k_t(\gamma_0)]$ and its asymptotic variance is independent of $\hat{\gamma}$, $\hat{\theta}$, and $\tilde{e}$. Because the effect of market-level variables in markets for which there is microdata persists in the limit, we use the notion of stable convergence. We say a sequence of random variables Z_T converges in distribution to Z ($\mathcal{M}$ -stably) if $\lim_{T \rightarrow \infty} \Pr(Z_T \leq z, A) = \Pr(Z \leq z, A)$ for all continuity points z of the distribution of Z and all $\mathcal{M}$ -measurable events A . Convergence in distribution is a special case corresponding to replacing $\mathcal{M}$ with the trivial σ -algebra $\{\emptyset, \Omega\}$.

Proposition 1. *Let Assumption 1 hold and $\hat{\gamma} = \gamma_0 + o_p(T^{-1/4})$. Then $\sqrt{T}(\hat{\kappa}_{bc} - \kappa_0)$ converges in distribution ($\mathcal{M}$ -stably) as $T \rightarrow \infty$ to a mixed Gaussian random variable with mean zero and $\mathcal{M}$ -measurable variance*

$$V_{bc} = \text{Var}(k_t(\gamma_0)) + h'H^{-1}h - K'V^{-1}K. \quad (27)$$

The (random) asymptotic variance V_{bc} can easily be estimated using $\hat{V}_{bc}$ in equation (15). Standard errors $\text{s.e.}(\hat{\kappa}_{bc}) = \sqrt{\hat{V}_{bc}/T}$ are consistent under Assumption 1 and valid inference can be performed based on t -statistics $(\hat{\kappa}_{bc} - \kappa_0)/\text{s.e.}(\hat{\kappa}_{bc})$ using the standard $N(0, 1)$ critical values.

Remark 9. Proposition 1 requires $\|\hat{\gamma} - \gamma_0\| = o_p(T^{-1/4})$, which is a standard condition used in asymptotics for plug-in estimators (e.g., Newey, 1994, Assumption 5.1(ii)). Of course, asymptotics are only useful insofar as they deliver accurate approximations to the finite-sample distribution of $\hat{\kappa}_{bc}$ encountered in practice. In practical terms, in any finite sample, this condition requires that $\hat{\gamma}$ is within a vicinity of γ_0 that is roughly double the order of sampling uncertainty. Indeed, the simulations reported in Section 4.1 show that $\hat{\kappa}_{bc}$ has negligible bias up to moderate amounts of error in $\tilde{e}$ (which translates to a range of $\|\hat{\gamma} - \gamma_0\|$) for a fixed sample size T . We also note that $\hat{\gamma} = \gamma(\hat{\theta}, \tilde{e})$, where $\hat{\theta}$ (and possibly $\tilde{e}$, via fine-tuning) is estimated on the choice data. For these reasons, it is plausible to adopt an asymptotic framework in which $\hat{\gamma}$ approaches γ_0 as the sample size T increases.

We next provide a sense in which $\hat{\kappa}_{bc}$ is efficient. The proof of Proposition 1 shows that $\hat{\kappa}_{bc}$ belongs to the class $\mathcal{K}$ of estimators $\hat{\kappa}$ of κ_0 that satisfy

$$\begin{aligned} \sqrt{T}(\hat{\kappa} - \kappa_0) &= \frac{1}{\sqrt{T}} \sum_{t=1}^T \left(k_t(\gamma_0) - \kappa_0 - c' Z_t \hat{\xi}_t(\gamma_0) \right) \\ &\quad + \sqrt{T} \sum_{t=1}^{\tau} d_t' (\bar{m}_t - m_t(\gamma_0)) + o_p(1), \end{aligned} \quad (28)$$

where c and $d_1, \dots, d_{\tau}$ are any $\mathcal{M}$ -measurable random vectors that satisfy

$$\mathbb{E}[\dot{k}_t(\gamma_0)] - G'c - \sum_{t=1}^{\tau} M_t' d_t = 0, \quad (29)$$

(almost surely). For instance, similar arguments as in the proof of Proposition 1 show that any $\hat{\kappa}$ obtained by plugging-in any $\hat{\gamma} = \gamma_0 + o_p(T^{-1/4})$ into (11) for some arbitrary weights $\hat{c}$ and $\hat{d}_1, \dots, \hat{d}_{\tau}$ converging to c and $d_1, \dots, d_{\tau}$ belongs to this class. Condition (29) typically ensures that such an estimator satisfies (28) uniformly for γ local to γ_0 . In general, there are many different weights $\hat{c}$ and $\hat{d}_1, \dots, \hat{d}_{\tau}$ whose probability limits c and $d_1, \dots, d_{\tau}$ will correspond to different (random) asymptotic

variances. The following result shows that $\hat{\kappa}_{bc}$ has the smallest asymptotic variance among this class of estimators of κ_0 .

Proposition 2. *Let Assumption 1 hold and $\hat{\gamma} = \gamma_0 + o_p(T^{-1/4})$. Then $\hat{\kappa}_{bc}$ has the smallest asymptotic variance among the class of estimators $\mathcal{K}$ of κ_0 .*

An important practical take-away from Proposition 2 is that microdata should be used, when available, to improve the efficiency of estimators of counterfactuals. Any estimator that discards the microdata by implicitly setting $d_1, \dots, d_\tau = 0$ will have an unnecessarily large variance (and hence standard errors).

Of course, in many scenarios microdata may not be available. Here we state a simpler version of Proposition 1 tailored to this case. Recall that the bias-corrected estimator in this case is given in (10), where $\hat{c}$ is given in (13) with $\hat{H} = \hat{G}'\hat{V}^{-1}\hat{G}$. To introduce the assumptions, let V and G be as above, and let $H = G'V^{-1}G$.

Assumption 2. Let the following hold:

- (i) $k_t(\cdot)$ and $Z_t\hat{\xi}_t(\cdot)$ are twice continuously differentiable in γ on N (almost surely), and the functions and all elements of their first and second derivatives are uniformly (for $\gamma \in N$) bounded by a random variable D_t with finite fourth moment;
- (ii) V and H are positive definite.

The next result is a special case of Proposition 1 and is stated without proof.

Proposition 3. *Let Assumption 2 hold and $\hat{\gamma} = \gamma_0 + o_p(T^{-1/4})$. Then,*

$$\sqrt{T}(\hat{\kappa}_{bc} - \kappa_0) \rightarrow_d N(0, V_{bc})$$

with V_{bc} as in (27) with $H = G'V^{-1}G$.

The asymptotic variance can be easily estimated using the formula $\hat{V}_{bc}$ in (16). Standard errors are then computed as $\sqrt{\hat{V}_{bc}/T}$. These are consistent under the conditions of Proposition 3.

6.1.2 Theory for LM_1

We now show that the diagnostic LM_1 in (17) behaves like $\|\sqrt{T}(\hat{\gamma} - \gamma_0)\|^2$ as the sample size grows large. In what follows, we abbreviate “with probability approaching one” to “wpa1.” Recall H from (26) and let $\lambda_{\min}(H)$ denote its smallest singular value, which is uniformly bounded away from zero by Assumption 1(iii).

Proposition 4. *Let Assumption 1 hold and let $\hat{\gamma} = \gamma_0 + o_p(1)$. Fix any sequence $C_T \uparrow \infty$ and any $\epsilon > 0$. Then wpa1, we have*

$$\frac{1 + \epsilon}{1 + 2\epsilon} \left(\sqrt{LM_1} - \epsilon C_T \right) \leq \|H^{1/2}(\sqrt{T}(\hat{\gamma} - \gamma_0))\| \leq \frac{1 + 2\epsilon}{1 + \epsilon} \left(\sqrt{LM_1} + \epsilon C_T \right).$$

In particular, wpa1 we have that $LM_1 \leq C_T^2$ implies

$$\|\sqrt{T}(\hat{\gamma} - \gamma_0)\| \leq \frac{(1 + 2\epsilon)C_T}{\sqrt{\lambda_{\min}(H)}}.$$

Moreover, if $\hat{\gamma} = \gamma_0 + o_p(C_T/\sqrt{T})$, then wpa1 we have

$$LM_1 \leq (1 + \epsilon)^2 C_T^2.$$

Proposition 4 shows LM_1 behaves like $\|\sqrt{T}(\hat{\gamma} - \gamma_0)\|^2$. With $C_T = o(T^{1/4})$, wpa1 we have that $LM_1 \leq C_T^2$ implies $\|\hat{\gamma} - \gamma_0\| \leq \text{constant} \times C_T/\sqrt{T} = o(T^{-1/4})$.

The proof of Proposition 4 shows that the “wpa1” qualifier depends on whether a $\chi_{\dim(\gamma)}^2$ random variable is less than $\epsilon^2 C_T^2$. With $\epsilon = 1$, say, this suggests taking C_T^2 to be at least as large as the 95th or 99th percentile of the $\chi_{\dim(\gamma)}^2$ distribution. To check a convergence rate of $\sqrt{(\log T)/T}$, for instance, one could use $C_T^2 = \chi_{\dim(\gamma), 0.95}^2 \log T$.

6.2 Case 2: Individual-Level Price Variation and Product Fixed Effects

We again let Γ denote the set of all values of $\gamma(\theta, e)$ as θ varies over the parameter space Θ and e varies over all $J \times r$ matrices whose first r rows are linearly independent, assume e_0 and $\tilde{e}$ are $J \times r$ and their first r rows are linearly independent, and that Γ is convex and open. This is true for the $\gamma(\theta, e)$ given in Example 3, for which Γ is the product of copies of $\mathbb{R}$ and $(0, \infty)$ and is therefore convex and open (see

footnote 22). In Section 3, the counterfactual function k_i and choice probabilities σ_i depended on (θ, e) only through the value of $\gamma(\theta, e)$. We therefore treat k_i and σ_i as random functions on Γ .

6.2.1 Theory for Bias Correction

We now derive the theoretical properties of the bias-corrected estimator $\hat{\kappa}_{bc}$ from (20). We first outline some standard smoothness, moment, and rank assumptions. In what follows, we use a dot (as above) and double dot to denote first and second derivatives with respect to γ . Let $H = \mathbb{E}[\dot{\sigma}_i(\gamma_0)' V_i(\gamma_0)^{-1} \dot{\sigma}_i(\gamma_0)]$ and N be a neighborhood of γ_0 .

Assumption 3. Let the following hold:

- (i) $k_i(\cdot)$ and $\sigma_i(\cdot)$ are twice continuously differentiable in γ on N (almost surely), and all elements of the functions and their first and second derivatives are uniformly (for $\gamma \in N$) bounded by a random variable D_i with finite second moment;
- (ii) $\dot{\sigma}_i(\cdot)' V_i(\cdot)^{-1}$ is continuously differentiable (almost surely) and all elements of the function and its derivative are uniformly (for $\gamma \in N$) bounded by a random variable D_i with finite higher-than-second moment;
- (iii) H is positive definite.

Let $\kappa_0 = \mathbb{E}[k_i(\gamma_0)]$ denote the true value of the counterfactual. The following result shows that the asymptotic distribution of $\hat{\kappa}_{bc}$ is centered at κ_0 and its variance is independent of $\hat{\gamma}$, $\hat{\theta}$, and $\tilde{e}$.

Proposition 5. *Let Assumption 3 hold and $\hat{\gamma} = \gamma_0 + o_p(n^{-1/4})$. Then*

$$\sqrt{n}(\hat{\kappa}_{bc} - \kappa_0) \rightarrow_d N(0, V_{bc}),$$

as $n \rightarrow \infty$, where

$$V_{bc} = \text{Var}(k_i(\gamma_0)) + \mathbb{E}[\dot{k}_i(\gamma_0)]' H^{-1} \mathbb{E}[\dot{k}_i(\gamma_0)]. \quad (30)$$

The asymptotic variance V_{bc} can be easily estimated using $\hat{V}_{bc}$ in (22). Standard errors are then computed as $\sqrt{\hat{V}_{bc}/n}$. These are consistent under the conditions of Proposition 5. We note that, as before, Proposition 5 requires that $\hat{\gamma}$ be in a vicinity of γ_0 , and refer the reader to Remark 9 for a discussion.

6.2.2 Theory for LM_1

The following result is analogous to Proposition 4 and shows LM_1 behaves like $\|\sqrt{n}(\hat{\gamma} - \gamma_0)\|^2$.

Proposition 6. *Let Assumption 3 hold and let $\hat{\gamma} = \gamma_0 + o_p(1)$. Fix any sequence $C_n \uparrow \infty$ and any $\epsilon > 0$. Then wpa1, we have*

$$\frac{1 + \epsilon}{1 + 2\epsilon} \left(\sqrt{LM_1} - \epsilon C_n \right) \leq \|H^{1/2}(\sqrt{n}(\hat{\gamma} - \gamma_0))\| \leq \frac{1 + 2\epsilon}{1 + \epsilon} \left(\sqrt{LM_1} + \epsilon C_n \right).$$

In particular, wpa1 we have that $LM_1 \leq C_n^2$ implies

$$\|\sqrt{n}(\hat{\gamma} - \gamma_0)\| \leq \frac{(1 + 2\epsilon)C_n}{\sqrt{\lambda_{\min}(H)}}.$$

Moreover, if $\hat{\gamma} = \gamma_0 + o_p(C_n/\sqrt{n})$, then wpa1 we have

$$LM_1 \leq (1 + \epsilon)^2 C_n^2.$$

The implications of Proposition 6 are similar to before. For $C_n = o(n^{1/4})$, we have wpa1 that $LM_1 \leq C_n^2$ implies $\|\hat{\gamma} - \gamma_0\| \leq \text{constant} \times C_n/\sqrt{n} = o(n^{-1/4})$. The proof of Proposition 6 shows that the “wpa1” qualifier depends on whether a $\chi_{\dim(\gamma)}^2$ random variable is less than $\epsilon^2 C_n^2$. To check a convergence rate of $\sqrt{(\log n)/n}$, for instance, one could use something like $C_n^2 = \chi_{\dim(\gamma), 0.95}^2 \log n$.

With some additional structure, we can also use LM_1 to deduce a similar bound on the proxies $\tilde{e}$. Let θ_* and e_* be such that $\gamma(\theta_*, e_*) = \gamma_0$. We do not require that θ_* and e_* are the true structural parameters and attributes, only that they induce γ_0 . Let $\hat{G}_\theta = \frac{\partial \gamma(\hat{\theta}, \tilde{e})'}{\partial \theta}$, $\hat{G}_e = \frac{\partial \gamma(\hat{\theta}, \tilde{e})'}{\partial \text{vec}(e)}$, $G_\theta = \frac{\partial \gamma(\theta_*, e_*)'}{\partial \theta}$, and $G_e = \frac{\partial \gamma(\theta_*, e_*)'}{\partial \text{vec}(e)}$ (these are well defined under Assumption 4 below). Also let $C(G_\theta)$ denote the column span of G_θ and $M = I - H^{1/2}G'_\theta(G_\theta H G'_\theta)^{-1}G_\theta H^{1/2}$ denote the projection onto $C(G_\theta)^\perp$.

Assumption 4. Let the following hold:

- (i) $\hat{\theta} \rightarrow_p \theta_*$ and $\tilde{e} \rightarrow_p e_*$ with $\gamma_0 = \gamma(\theta_*, e_*)$;
- (ii) $\gamma(\theta, e)$ is continuously differentiable in both its arguments at (θ_*, e_*) and G_θ has full row rank;
- (iii) $\hat{\theta}$ satisfies the first-order condition $0 = \hat{G}_\theta \hat{S}$ and there exists a constant C such that $\|\hat{\theta} - \theta_*\| \leq C\|\tilde{e} - e_*\|$ wpa1;

(iv) $MH^{1/2}G'_e$ has full rank.

Let $\sigma_{\min}(MH^{1/2}G'_e)$ denote the smallest singular value of the matrix $MH^{1/2}G'_e$. Note this is positive by Assumption 4(iv).

Proposition 7. *Let Assumptions 3 and 4 hold. Fix any sequence $C_n \uparrow \infty$ and any $\epsilon > 0$. Then wpa1, we have*

$$\frac{1 + \epsilon}{1 + 3\epsilon} \left(\sqrt{LM_1} - \epsilon C_n \right) \leq \|MH^{1/2}G'_e \sqrt{n}(\text{vec}(\tilde{e} - e_*))\| \leq \frac{1 + 3\epsilon}{1 + \epsilon} \left(\sqrt{LM_1} + \epsilon C_n \right).$$

In particular, wpa1 we have that $LM_1 \leq C_n^2$ implies

$$\|\sqrt{n}(\text{vec}(\tilde{e} - e_*))\| \leq \frac{(1 + 3\epsilon)C_n}{\sigma_{\min}(MH^{1/2}G'_e)}.$$

The proof of Proposition 7 shows that the “wpa1” qualifier depends on whether a $\chi^2_{\text{rank}(M)}$ random variable is less than $\epsilon^2 C_n^2$. With $\epsilon = 1$, say, this suggests taking C_n^2 to be at least as large as the 95th or 99th percentile of the $\chi^2_{\text{rank}(M)}$ distribution.

7 Conclusion

In this paper, we develop a toolkit to correct bias and perform valid inference on counterfactuals when the product attributes used in demand estimation may only imperfectly capture the latent attributes that drive substitution. A leading case is when consumer choice is driven by difficult-to-quantify characteristics and unstructured data, such as product images, descriptions, review text, or consumer surveys, are converted into numerical variables using ML methods. As e-commerce continues to expand and such data play an increasingly central role in driving consumer choices, the need to incorporate these sources into demand estimation will only grow. In addition, our methods may be applied as simple post-estimation robustness checks even with standard numeric attributes when they are imperfect proxies. All our methods require minimal additional computation once model parameters are estimated and can be easily integrated in the canonical demand estimation workflow.

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A Proofs

A.1 Proofs for Section 6.1

Proof of Proposition 1. We have $\hat{\gamma} \in N$ wpa1 by the assumed consistency of $\hat{\gamma}$. By Assumption 1(i), wpa1 we may take a mean value expansion around γ_0 to obtain

$$\begin{aligned} \sqrt{T}(\hat{\kappa}_{bc} - \kappa_0) &= \frac{1}{\sqrt{T}} \sum_{t=1}^T k_t(\gamma_0) - \kappa_0 - c' Z_t \hat{\xi}_t(\gamma_0) + \sqrt{T} \sum_{t=1}^{\tau} d'_t (\bar{m}_t - m_t(\gamma_0)) \\ &\quad + \frac{1}{\sqrt{T}} \sum_{t=1}^T \left((\hat{c} - c)' Z_t \hat{\xi}_t(\gamma_0) + \sum_{s=1}^{\tau} (\hat{d}_s - d_s)' (\bar{m}_s - m_s(\gamma_0)) \right) \\ &\quad + \frac{1}{\sqrt{T}} \sum_{t=1}^T \left(\dot{k}_t(\tilde{\gamma})' - \dot{c}' Z_t \dot{\xi}_t(\tilde{\gamma}) - \sum_{s=1}^{\tau} \dot{d}'_s \dot{m}_s(\tilde{\gamma}) \right) (\hat{\gamma} - \gamma_0) \\ &=: T_{1,T} + T_{2,T} + T_{3,T}, \end{aligned}$$

where $\tilde{\gamma}$ is in the segment between $\hat{\gamma}$ and γ_0 , and

$$c = V^{-1}(K + GH^{-1}h), \quad d_t = (r_t V_t)^{-1} M_t H^{-1} h, \quad t = 1, \dots, \tau. \quad (31)$$

Note that c and $d_1, \dots, d_{\tau}$ are well defined by virtue of Assumption 1(iii).

For $T_{1,T}$, define the $(\dim(z) + 1) \times 1$ random vector $\zeta_t = (k_t(\gamma_0) - \kappa_0, Z_t \hat{\xi}_t(\gamma_0))$. By Theorem 2 of [Hahn et al. \(2022\)](#) (noting Assumptions 1(i)(ii) and independence within and across markets are sufficient for their integrability and dependence conditions) and Assumption 1(iv), for any $\mathcal{M}$ -measurable random vectors $d_1, \dots, d_{\tau}$, we have

$$\left(\begin{array}{c} \frac{1}{\sqrt{T}} \sum_{t=1}^T \zeta_t \\ \sum_{t=1}^{\tau} d'_t \sqrt{T} (\bar{m}_t - m_t(\gamma_0)) \end{array} \right) \rightarrow_d \left(\begin{array}{c} Z_A \\ (\sum_{t=1}^{\tau} r_t d'_t V_t d_t)^{1/2} Z_M \end{array} \right)$$

$\mathcal{M}$ -stably, where the random vector Z_A and random variable Z_M are jointly normally distributed and independent with mean zero, $\text{Var}(Z_A) = \text{Var}(\zeta_t)$, and $\text{Var}(Z_M) = 1$. Moreover, (Z_A, Z_M) are independent of any $\mathcal{M}$ -measurable random variable. Hence, the asymptotic distribution of $T_{1,T}$ is mixed Gaussian with mean zero and random variance

$$\text{Var}(k_t(\gamma_0)) + c' V c - 2c' K + \sum_{t=1}^{\tau} r_t d'_t V_t d_t.$$

Substituting the above formulas for c and $d_1, \dots, d_\tau$ gives the form of the variance in display (27). It remains to show that $T_{2,T}$ and $T_{3,T}$ are asymptotically negligible.

For term $T_{2,T}$, first recall the expressions for $\hat{h}$ and $\hat{H}$ in (12). Note that by Assumption 1(i)(ii) and consistency of $\hat{\gamma}$, we can deduce by standard arguments (e.g., Lemma 2.4 of Newey and McFadden (1994)) that $\hat{k} \rightarrow_p \mathbb{E}[\dot{k}_t(\gamma_0)]$, $\hat{K} \rightarrow_p K$, $\hat{G} \rightarrow_p G$, and $\hat{V} \rightarrow_p V$. Hence, $\hat{h} \rightarrow_p h$ by Assumption 1(iii) and Slutsky's theorem. Note that for each $1 \leq t \leq \tau$, the m_{it} are iid conditional on $\mathcal{M}$. It follows by Lemma 1 of Andrews (2005) and Assumption 1(ii)(iv) that $\hat{V}_t \rightarrow_p r_t V_t$. Hence, $\hat{V}_t^{-1} \rightarrow_p (r_t V_t)^{-1}$ for $1 \leq t \leq \tau$ by Assumption 1(iii). Finally, Assumption 1(i) implies $\hat{M}_t \rightarrow_p M_t$ for $1 \leq t \leq \tau$. Hence, $\hat{H} \rightarrow_p H$ and so $\hat{H}^{-1} \rightarrow_p H^{-1}$, $\hat{c} \rightarrow_p c$, and $\hat{d}_t \rightarrow_p d_t$ for $1 \leq t \leq \tau$ by Assumption 1(iii) and Slutsky's theorem.

Now write

$$T_{2,T} = (\hat{c} - c)' \frac{1}{\sqrt{T}} \sum_{t=1}^T Z_t \hat{\xi}_t(\gamma_0) + \sum_{t=1}^{\tau} \frac{\sqrt{T}}{\sqrt{N_t}} (\hat{d}_t - d_t)' \sqrt{N_t} (\bar{m}_t - m_t(\gamma_0))$$

$$=: T_{2,T,a} + T_{2,T,b}.$$

Term $T_{2,T,a} \rightarrow_p 0$ because $\hat{c} \rightarrow_p c$ and $\frac{1}{\sqrt{T}} \sum_{t=1}^T Z_t \hat{\xi}_t(\gamma_0) = O_p(1)$ by Assumption 1(i). Similarly, $T_{2,T,b} \rightarrow_p 0$ because $\sqrt{T/N_t} \rightarrow r_t \in (0, \infty)$ by Assumption 1(iv), $\hat{d}_t \rightarrow_p d_t$, and $\sqrt{N_t}(\bar{m}_t - m_t(\gamma_0))$ converges in distribution $\mathcal{M}$ -stably to a mixed normal limit with mean zero and variance V_t (by Assumption 1(ii)) and is therefore tight by Assumption 1(ii).

For term $T_{3,T}$, we first let $m_{tl}(\gamma)$ denote the l -th element of $m_t(\gamma)$, and let $\rho_{tl}(\gamma)$ denote the l -th element of $Z_t \hat{\xi}_t(\gamma)$. Similarly, we let $\hat{c}_l$ and $\hat{d}_{tl}$ denote the l -th elements of $\hat{c}$ and $\hat{d}_t$. Then we may write

$$T_{3,T} = \frac{1}{\sqrt{T}} \sum_{t=1}^{\tau} \left(\dot{k}_t(\tilde{\gamma})' - \sum_{l=1}^{\dim(z)} \hat{c}_l \dot{\rho}_{tl}(\tilde{\gamma})' - \sum_{s=1}^{\tau} \sum_{l=1}^{\dim(m)} \hat{d}_{sl} \dot{m}_{sl}(\tilde{\gamma})' \right) (\hat{\gamma} - \gamma_0).$$

By construction, $\hat{c}$ and $\hat{d}_1, \dots, \hat{d}_\tau$ satisfy the in-sample orthogonality condition

$$\hat{k} - \hat{G}' \hat{c} - \sum_{s=1}^{\tau} \hat{M}_s' \hat{d}_s = 0, \quad (32)$$

wpa1. By Assumption 1(i) we may take a second mean-value expansion, this time of

$\tilde{\gamma}$ around $\hat{\gamma}$, to arrive at

$$\begin{aligned}
T_{3,T} &= \frac{1}{\sqrt{T}} \sum_{t=1}^T \left(\dot{k}_t(\hat{\gamma})' - \sum_{l=1}^{\dim(z)} \hat{c}_l \dot{\rho}_{tl}(\hat{\gamma})' - \sum_{s=1}^{\tau} \sum_{l=1}^{\dim(m)} \hat{d}_{sl} \dot{m}_{sl}(\hat{\gamma})' \right) (\hat{\gamma} - \gamma_0) \\
&\quad + T^{1/4}(\tilde{\gamma} - \hat{\gamma})' \left(\frac{1}{T} \sum_{t=1}^T \left(\ddot{k}_t(\tilde{\gamma}) - \sum_{l=1}^{\dim(z)} \hat{c}_l \ddot{\rho}_{tl}(\tilde{\gamma}) - \sum_{s=1}^{\tau} \sum_{l=1}^{\dim(m)} \hat{d}_{sl} \ddot{m}_{sl}(\tilde{\gamma}) \right) \right) T^{1/4}(\hat{\gamma} - \gamma_0) \\
&=: T_{3,T,a} + T_{3,T,b},
\end{aligned}$$

wpa1, where $\tilde{\gamma}$ is in the segment between $\hat{\gamma}$ and γ_0 , $\ddot{k}_t(\gamma) = \frac{\partial^2 k_t(\gamma)}{\partial \gamma \partial \gamma'}$, $\ddot{\rho}_{tl}(\gamma) = \frac{\partial^2 \rho_{tl}(\gamma)}{\partial \gamma \partial \gamma'}$, and $\ddot{m}_{sl}(\gamma) = \frac{\partial^2 m_{sl}(\gamma)}{\partial \gamma \partial \gamma'}$.

We have

$$T_{3,T,a} = \left(\hat{k} - \hat{G}' \hat{c} - \sum_{s=1}^{\tau} \hat{M}'_s \hat{d}_s \right) \sqrt{T}(\hat{\gamma} - \gamma_0) = 0$$

wpa1 by the in-sample orthogonality condition (32).

To show $T_{3,T,b} \rightarrow_p 0$, in view of the condition $\hat{\gamma} = \gamma_0 + o_p(T^{-1/4})$, it is enough to show that the central term in parentheses is $O_p(1)$. To this end, standard arguments (e.g., Lemma 2.4 of Newey and McFadden (1994)) using Assumption 1(i) and consistency of $\hat{\gamma}$ yield $\frac{1}{T} \sum_{t=1}^T \ddot{k}_t(\tilde{\gamma}) \rightarrow_p \mathbb{E}[\ddot{k}_t(\gamma_0)]$ and $\frac{1}{T} \sum_{t=1}^T \ddot{\rho}_{tl}(\tilde{\gamma}) \rightarrow_p \mathbb{E}[\ddot{\rho}_{tl}(\gamma_0)]$, both of which are finite. It also follows by the fact that $\hat{d}_t \rightarrow_p d_t$ for $1 \leq t \leq \tau$, Assumption 1(i), and consistency of $\hat{\gamma}$ that $\hat{d}_{sl} \ddot{m}_{sl}(\tilde{\gamma}) \rightarrow_p d_{sl} \ddot{m}_{sl}(\gamma_0)$ for $1 \leq s \leq \tau$ and $1 \leq l \leq L$. Finally, Assumption 1(i)-(iii) implies c and $d_1, \dots, d_\tau$ are tight. $\square$

Proof of Proposition 2. Arguing as in the proof of Proposition 1, the asymptotic distribution of any estimator of the form (28) is mixed Gaussian with mean zero and variance

$$\text{Var}(k_t(\gamma_0)) + c' V c - 2c' K + \sum_{t=1}^{\tau} r_t d_t' V_t d_t.$$

Conditioning on $\mathcal{M}$, we may minimize this expression with respect to the vectors c and $d_1, \dots, d_\tau$ subject to (29) to obtain the weights c and $d_1, \dots, d_\tau$ in (31). Substituting into the above display yields the minimum variance V_{bc} given in (27). $\square$

Before proving Proposition 4, we first state and prove a lemma.

Lemma 1. *Let Assumption 1 hold and let $\hat{\gamma} = \gamma_0 + o_p(1)$. Then*

$$\sqrt{T}\hat{S} = Z_T + o_p(1) + (H + o_p(1))(\sqrt{T}(\hat{\gamma} - \gamma_0)),$$

where $Z_T := G'V^{-1}\frac{1}{\sqrt{T}}\sum_{t=1}^T Z_t\hat{\xi}_t(\gamma_0) + \sum_{t=1}^{\tau} M'_t(r_tV_t)^{-1}\sqrt{T}(m_t(\gamma_0) - \bar{m}_t)$ converges in distribution ($\mathcal{M}$ -stably) to a mixed Gaussian random variable with mean zero and $\mathcal{M}$ -measurable variance H .

Proof of Lemma 1. By definition of $\hat{S}$, we have

$$\begin{aligned}\sqrt{T}\hat{S} &= \hat{G}'\hat{V}^{-1}\frac{1}{\sqrt{T}}\sum_{t=1}^T Z_t\hat{\xi}_t(\gamma_0) + \sum_{t=1}^{\tau} \hat{M}'_t\hat{V}_t^{-1}\sqrt{T}(m_t(\gamma_0) - \bar{m}_t) \\ &\quad + \hat{G}'\hat{V}^{-1}\left(\frac{1}{\sqrt{T}}\sum_{t=1}^T Z_t(\hat{\xi}_t(\hat{\gamma}) - \hat{\xi}_t(\gamma_0))\right) \\ &\quad + \sum_{t=1}^{\tau} \hat{M}'_t\hat{V}_t^{-1}\sqrt{T}(m_t(\hat{\gamma}) - m_t(\gamma_0)) =: T_{1,T} + T_{2,T} + T_{3,T} + T_{4,T},\end{aligned}$$

where $\hat{G} \rightarrow_p G$, $\hat{M}_t \rightarrow_p M_t$ for $1 \leq t \leq \tau$, $\hat{V}^{-1} \rightarrow_p V^{-1}$, and $\hat{V}_t^{-1} \rightarrow_p (r_tV_t)^{-1}$ for $1 \leq t \leq \tau$ (all by the proof of Proposition 1).

For $T_{1,T}$ and $T_{2,T}$, we have by the proof of Proposition 1 that $\frac{1}{\sqrt{T}}\sum_{t=1}^T Z_t\hat{\xi}_t(\gamma_0)$ and $\sqrt{N_t}(m_t(\gamma_0) - \bar{m}_t)$, $1 \leq t \leq \tau$, are all $O_p(1)$. It follows by Assumption 1(iv) that $T_{1,T} + T_{2,T} = Z_T + o_p(1)$. Hence, by similar arguments to the proof of Proposition 1 we may invoke Theorem 2 of Hahn et al. (2022) to conclude that Z_T converges $\mathcal{M}$ -stably to a mixed Gaussian limit with mean zero and variance H .

For $T_{3,T}$ and $T_{4,T}$, a mean-value expansion in $\hat{\gamma}$ around γ_0 yields

$$T_{3,T} = \hat{G}'\hat{V}^{-1}\left(\frac{1}{T}\sum_{t=1}^T Z_t\dot{\xi}_t(\tilde{\gamma})\right)\sqrt{T}(\hat{\gamma} - \gamma_0), \quad T_{4,T} = \sum_{t=1}^{\tau} \hat{M}'_t\hat{V}_t^{-1}\dot{m}_t(\tilde{\gamma})'\sqrt{T}(\hat{\gamma} - \gamma_0),$$

for $\tilde{\gamma}$ in the segment between $\hat{\gamma}$ and γ_0 . It follows by Assumption 1(i) and standard arguments that $\frac{1}{T}\sum_{t=1}^T Z_t\dot{\xi}_t(\tilde{\gamma}) \rightarrow_p G$ and $\dot{m}_t(\tilde{\gamma}) \rightarrow_p M_t$, $1 \leq t \leq \tau$. Hence, $T_{3,T} + T_{4,T} = (H + o_p(1))\sqrt{T}(\hat{\gamma} - \gamma_0)$. $\square$

Proof of Proposition 4. The proof of Proposition 1 shows that $\hat{H} \rightarrow_p H$. Combined

with Lemma 1 and the triangle inequality, we have

$$\begin{aligned} & \| (H^{1/2} + o_p(1))(\sqrt{T}(\hat{\gamma} - \gamma_0)) \| + \| (H^{-1/2} + o_p(1))(Z_T + o_p(1)) \| \\ & \geq \sqrt{LM_1} \geq \| (H^{1/2} + o_p(1))(\sqrt{T}(\hat{\gamma} - \gamma_0)) \| - \| (H^{-1/2} + o_p(1))(Z_T + o_p(1)) \|. \end{aligned}$$

As $\| (H^{-1/2} + o_p(1))(Z_T + o_p(1)) \|^2 \rightarrow_d \chi_{\dim(\gamma)}^2$ by the proof of Lemma 1, we have

$$\| (H^{-1/2} + o_p(1))(Z_T + o_p(1)) \| \leq \epsilon C_T$$

wpa1. Moreover, we have that

$$\frac{1 + 2\epsilon}{1 + \epsilon} \| H^{1/2}(\sqrt{T}(\hat{\gamma} - \gamma_0)) \| \geq \| (H^{1/2} + o_p(1))(\sqrt{T}(\hat{\gamma} - \gamma_0)) \| \geq \frac{1 + \epsilon}{1 + 2\epsilon} \| H^{1/2}(\sqrt{T}(\hat{\gamma} - \gamma_0)) \|$$

wpa1. The first result follows by combining the above three displays and rearranging. The second and third results are implications of the first. $\square$

A.2 Proofs for Section 6.2

Proof of Proposition 5. Let $c'_i = (\mathbb{E}[\dot{k}_i(\gamma_0)])' H^{-1} \dot{\sigma}_i(\gamma_0)' V_i(\gamma_0)^{-1}$ denote the population counterpart of $\hat{c}'_i$. We have $\hat{\gamma} \in N$ wpa1 by consistency of $\hat{\gamma}$. Hence, wpa1, we may take a mean value expansion around γ_0 to obtain

$$\begin{aligned} \sqrt{n}(\hat{\kappa}_{bc} - \kappa_0) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n k_i(\gamma_0) - \kappa_0 + c'_i(d_i - \sigma_i(\gamma_0)) \\ &\quad + \frac{1}{\sqrt{n}} \sum_{i=1}^n (\hat{c}_i - c_i)'(d_i - \sigma_i(\gamma_0)) \\ &\quad + \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\dot{k}_i(\tilde{\gamma})' - \hat{c}'_i \dot{\sigma}_i(\tilde{\gamma}) \right) (\hat{\gamma} - \gamma_0) \\ &=: T_{1,n} + T_{2,n} + T_{3,n}, \end{aligned}$$

where $\tilde{\gamma}$ is in the segment between $\hat{\gamma}$ and γ_0 . This expansion is valid by Assumption 3(i). Term $T_{1,n}$ is asymptotically $N(0, V_{bc})$. It remains to show that $T_{2,n}$ and $T_{3,n}$ are both asymptotically negligible.

For $T_{2,n}$, first define the $1 \times \dim(\gamma)$ vectors

$$\hat{a} = \bar{k}' \hat{H}^{-1}, \quad a = \mathbb{E}[\dot{k}_i(\gamma_0)]' H^{-1},$$

where $\bar{k} = \frac{1}{n} \sum_{i=1}^n \dot{k}_i(\hat{\gamma})$. Also define the $\dim(\gamma) \times J$ random element $b_i(\gamma) = \dot{\sigma}_i(\gamma)' V_i(\gamma)^{-1}$, and let $e_i = d_i - \sigma_i(\gamma_0)$, which is $J \times 1$. Then we may write

$$\begin{aligned} T_{2,n} &= \hat{a} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n (b_i(\hat{\gamma}) - b_i(\gamma_0)) e_i \right) + (\hat{a} - a) \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n b_i(\gamma_0) e_i \right) \\ &=: T_{2,n,a} + T_{2,n,b}. \end{aligned}$$

By Assumptions 3(i)(ii) and consistency of $\hat{\gamma}$, it follows by standard arguments (e.g., Newey and McFadden, 1994, Lemma 2.4) that $\bar{k} \rightarrow_p \mathbb{E}[\dot{k}_i(\gamma_0)]$ and $\hat{H} \rightarrow_p H$. Hence, $\hat{a} \rightarrow_p a$ by Assumption 3(iii).

To show $T_{2,n,a} \rightarrow_p 0$, first note that $\hat{a} = O_p(1)$ and $\mathbb{E}[b_i(\gamma) e_i] = 0$. Consider the empirical process $\nu_n(\gamma) = \frac{1}{\sqrt{n}} \sum_{i=1}^n b_i(\gamma) e_i$ defined for $\gamma \in N$. For $\gamma_1, \gamma_2 \in N$, we have by a mean-value expansion that $b_i(\gamma_1) e_i - b_i(\gamma_2) e_i = (\gamma_1 - \gamma_2)' \dot{b}_i(\tilde{\gamma}) e_i$ for $\tilde{\gamma}$ in the segment between γ_1 and γ_2 (with possibly different values for each element), where $\dot{b}_i(\gamma) e_i = \frac{\partial}{\partial \gamma} (\dot{b}_i(\gamma) e_i)'$. This expansion is valid in view of Assumption 3(ii). The elements of e_i are bounded by ± 1 and the elements of $\dot{b}_i(\gamma)$ are uniformly (for $\gamma \in N$) bounded by some random variable with finite second moment, again by Assumption 3(ii). Hence, $\|b_i(\gamma_1) e_i - b_i(\gamma_2) e_i\| \leq B_i \|\gamma_1 - \gamma_2\|$ for $\gamma_1, \gamma_2 \in N$, for some random variable B_i with finite second moment. Thus, $\{b_i(\gamma) e_i : \gamma \in N\}$ is a type-II class of Andrews (1994b). It follows by Theorems 1 and 2 of Andrews (1994b) (using Assumption 3(ii) to verify the moment condition on the envelope function) that $\nu_n(\cdot)$ is stochastically equicontinuous. Also note by the Lipschitz condition the pseudometric corresponding to this process is dominated by the Euclidean metric. Hence, by consistency of $\hat{\gamma}$ we have $\frac{1}{\sqrt{n}} \sum_{i=1}^n (b_i(\hat{\gamma}) - b_i(\gamma_0)) e_i \rightarrow_p 0$.

To show $T_{2,n,b} \rightarrow_p 0$, first note $\hat{a} \rightarrow_p a$. Moreover, $\mathbb{E}[\|b_i(\gamma_0) e_i\|^2] < \infty$ by Assumption 3(ii) and $\mathbb{E}[b_i(\gamma_0) e_i] = 0$, so $\frac{1}{\sqrt{n}} \sum_{i=1}^n b_i(\gamma_0) e_i = O_p(1)$ by Chebyshev's inequality.

For term $T_{3,n}$, we first write

$$T_{3,n} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\dot{k}_i(\hat{\gamma})' - \sum_{j=1}^J \hat{c}_{ij} \dot{\sigma}_{ij}(\hat{\gamma})' \right) (\hat{\gamma} - \gamma_0),$$

where $\hat{c}_i = (\hat{c}_{i1}, \dots, \hat{c}_{iJ})$, and $\dot{\sigma}_{ij}(\gamma) = \frac{\partial \sigma_{ij}(\gamma)}{\partial \gamma}$. Note by construction that $\hat{c}_i$ satisfies the in-sample orthogonality condition

$$\frac{1}{n} \sum_{i=1}^n \dot{k}_i(\hat{\gamma}) - \dot{\sigma}_i(\hat{\gamma})' \hat{c}_i = 0. \quad (33)$$

A second mean-value expansion, this time of $\tilde{\gamma}$ around $\hat{\gamma}$, yields

$$\begin{aligned} T_{3,n} &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\dot{k}_i(\hat{\gamma})' - \hat{c}_i' \dot{\sigma}_i(\hat{\gamma}) \right) (\hat{\gamma} - \gamma_0) \\ &\quad + n^{1/4} (\tilde{\gamma} - \hat{\gamma})' \left(\frac{1}{n} \sum_{i=1}^n \left(\ddot{k}_i(\tilde{\gamma}) - \sum_{j=1}^J \hat{c}_{ij} \ddot{\sigma}_{ij}(\tilde{\gamma}) \right) \right) n^{1/4} (\hat{\gamma} - \gamma_0) \\ &=: T_{3,n,a} + T_{3,n,b}, \end{aligned}$$

where $\tilde{\gamma}$ is in the segment between $\hat{\gamma}$ and γ_0 , $\ddot{k}_i(\gamma) = \frac{\partial^2 k_i(\gamma)}{\partial \gamma \partial \gamma'}$, and $\ddot{\sigma}_{ij}(\gamma) = \frac{\partial^2 \sigma_{ij}(\gamma)}{\partial \gamma \partial \gamma'}$. We have $T_{3,n,a} = 0$ by the in-sample orthogonality condition (33).

To show $T_{3,n,b} \rightarrow_p 0$, in view of the condition $\hat{\gamma} = \gamma_0 + o_p(n^{-1/4})$, it is enough to show that the central term in parentheses is $O_p(1)$. To this end, standard arguments (e.g., Newey and McFadden, 1994, Lemma 2.4) using Assumption 3(i) and consistency of $\hat{\gamma}$ yield $\frac{1}{n} \sum_{i=1}^n \ddot{k}_i(\tilde{\gamma}) \rightarrow_p \mathbb{E}[\ddot{k}_i(\gamma_0)]$, which is finite. We may similarly deduce by the fact that $\hat{a} \rightarrow_p a$ and Assumption 3(i)-(iii) that $\frac{1}{n} \sum_{i=1}^n \hat{c}_{ij} \ddot{\sigma}_{ij}(\tilde{\gamma}) \rightarrow_p \mathbb{E}[c_{ij} \ddot{\sigma}_{ij}(\gamma_0)]$, which is finite, for $j = 1, \dots, J$. $\square$

Proof of Proposition 6. Analogous to the proof of Proposition 4, using Lemma 2 below in place of Lemma 1. $\square$

Lemma 2. *Let Assumption 3 hold and let $\hat{\gamma} = \gamma_0 + o_p(1)$. Then*

$$\sqrt{n} \hat{S} = Z_n + o_p(1) - (H + o_p(1))(\sqrt{n}(\hat{\gamma} - \gamma_0)),$$

where $Z_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n \dot{\sigma}_i(\gamma_0)' V_i(\gamma_0)^{-1} (d_i - \sigma_i(\gamma_0)) \rightarrow_d N(0, H)$.

Proof of Lemma 2. First note that since $\sum_{j=0}^J \dot{\sigma}_{ij}(\gamma) = 0$ for all $\gamma \in \Gamma$, we may write

$$\begin{aligned} \sqrt{n} \hat{S} &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \sum_{j=0}^J \frac{d_{ij} - \sigma_{ij}(\gamma_0)}{\sigma_{ij}(\hat{\gamma})} \dot{\sigma}_{ij}(\hat{\gamma}) + \frac{1}{\sqrt{n}} \sum_{i=1}^n \sum_{j=0}^J \frac{\sigma_{ij}(\gamma_0) - \sigma_{ij}(\hat{\gamma})}{\sigma_{ij}(\hat{\gamma})} \dot{\sigma}_{ij}(\hat{\gamma}) \\ &=: T_{1,n} + T_{2,n}. \end{aligned}$$

For $T_{1,n}$, we may rewrite this term using the notation from the proof of Proposition 5 as

$$\begin{aligned} T_{1,n} &= \frac{1}{\sqrt{n}} \sum_{i=1}^n b_i(\gamma_0) e_i + \frac{1}{\sqrt{n}} \sum_{i=1}^n (b_i(\hat{\gamma}) - b_i(\gamma_0)) e_i \\ &=: T_{1,n,a} + T_{1,n,b} \end{aligned}$$

where $b_i(\gamma) = \dot{\sigma}_i(\gamma)' V_i(\gamma)^{-1}$ and $e_i = d_i - \sigma_i(\gamma_0)$. The summands in $T_{1,n,a}$ have mean zero and variance H , which is finite and non-singular by Assumption 3(iii). Hence, $T_{1,n,a} \rightarrow_d N(0, H)$. The proof of Proposition 5 shows that the empirical process $\nu_n(\gamma) = \frac{1}{\sqrt{n}} \sum_{i=1}^n b_i(\gamma) e_i$ defined for $\gamma \in N$, a suitable neighborhood of γ_0 , is stochastically equicontinuous under Assumption 3(ii). Hence, $T_{1,n,b} \rightarrow_p 0$.

For $T_{2,n}$, a mean-value expansion in γ_0 around $\hat{\gamma}$ yields

$$T_{2,n} = \left(\frac{1}{n} \sum_{i=1}^n \sum_{j=0}^J \frac{\dot{\sigma}_{ij}(\hat{\gamma}) \dot{\sigma}_{ij}(\tilde{\gamma})'}{\sigma_{ij}(\hat{\gamma})} \right) \sqrt{n}(\gamma_0 - \hat{\gamma}),$$

for $\tilde{\gamma}$ in the segment between γ_0 and $\hat{\gamma}$ (with possibly different values for each element). This expansion is valid in view of Assumption 3(i). For the term in parentheses, note

$$\frac{1}{n} \sum_{i=1}^n \sum_{j=0}^J \frac{\dot{\sigma}_{ij}(\hat{\gamma}) \dot{\sigma}_{ij}(\tilde{\gamma})'}{\sigma_{ij}(\hat{\gamma})} = \frac{1}{n} \sum_{i=1}^n \dot{\sigma}_i(\hat{\gamma})' V_i(\hat{\gamma})^{-1} \dot{\sigma}_i(\tilde{\gamma}).$$

Standard arguments (e.g., Lemma 2.4 of Newey and McFadden (1994)) then yield that $\frac{1}{n} \sum_{i=1}^n \dot{\sigma}_i(\hat{\gamma})' V_i(\hat{\gamma})^{-1} \dot{\sigma}_i(\tilde{\gamma}) \rightarrow_p H$ under Assumption 3(i)(ii). $\square$

Proof of Proposition 7. First note that Assumption 4(i)(ii) implies $\hat{\gamma} \rightarrow_p \gamma_0$. Moreover, $\hat{H} \rightarrow_p H$ by the proof of Proposition 5, H is positive definite by Assumption 3(iii), and $\hat{G}_\theta \rightarrow_p G_\theta$ by Assumption 4(i)(ii). It follows by Assumption 4(ii) that $\hat{M} = I - \hat{H}^{1/2} \hat{G}_\theta' (\hat{G}_\theta \hat{H} \hat{G}_\theta')^{-1} \hat{G}_\theta \hat{H}^{1/2}$ exists wpa1 and $\hat{M} \rightarrow_p M$. Now by Assumption 4(iii) and Lemma 2,

$$\sqrt{n} \hat{S} = \sqrt{n} \hat{M} \hat{H}^{-1/2} \hat{S} = \hat{M} \hat{H}^{-1/2} (Z_n + o_p(1)) - \hat{M} \hat{H}^{-1/2} (H + o_p(1)) (\sqrt{n}(\hat{\gamma} - \gamma_0)),$$

where $\hat{M} = I - \hat{H}^{1/2} \hat{G}_\theta' (\hat{G}_\theta \hat{H} \hat{G}_\theta')^{-1} \hat{G}_\theta \hat{H}^{1/2}$. Hence,

$$\sqrt{n} \hat{S} = (M + o_p(1)) (H^{-1/2} Z_n + o_p(1)) - (M H^{1/2} + o_p(1)) (\sqrt{n}(\hat{\gamma} - \gamma_0)).$$

A mean-value expansion of $\hat{\gamma}$ around (θ_*, e_*) yields

$$\sqrt{n}(\hat{\gamma} - \gamma_0) = (G'_\theta + o_p(1))\sqrt{n}(\hat{\theta} - \theta_*) + (G'_e + o_p(1))\sqrt{n}(\text{vec}(\tilde{e} - e_*)).$$

Since $MH^{1/2}G'_\theta = 0$, we have by Assumption 4(iii)(iv) that

$$\|(MH^{1/2} + o_p(1))(G'_\theta + o_p(1))\sqrt{n}(\hat{\theta} - \theta_*)\| \leq \frac{\epsilon}{1 + 3\epsilon} \|MH^{1/2}G'_e\sqrt{n}(\text{vec}(\tilde{e} - e_*))\|$$

wpa1. Moreover,

$$\begin{aligned} \frac{1 + 2\epsilon}{1 + 3\epsilon} \|MH^{1/2}G'_e\sqrt{n}(\text{vec}(\tilde{e} - e_*))\| \\ \leq \|(MH^{1/2} + o_p(1))(G'_e + o_p(1))\sqrt{n}(\text{vec}(\tilde{e} - e_*))\| \\ \leq \frac{1 + 2\epsilon}{1 + \epsilon} \|MH^{1/2}G'_e\sqrt{n}(\text{vec}(\tilde{e} - e_*))\| \end{aligned}$$

wpa1. We also have $\|MH^{-1/2}Z_n\|^2 \rightarrow_d \chi^2_{\text{rank}(M)}$ by Lemma 2, which implies that the inequality $\|(M + o_p(1))(H^{-1/2}Z_n + o_p(1))\| \leq \epsilon C_n$ holds wpa1. Hence, wpa1,

$$\begin{aligned} \frac{1 + 3\epsilon}{1 + \epsilon} \|MH^{1/2}G'_e\sqrt{n}(\text{vec}(\tilde{e} - e_*))\| + \epsilon C_n \\ \geq \sqrt{LM_1} \geq \frac{1 + \epsilon}{1 + 3\epsilon} \|MH^{1/2}G'_e\sqrt{n}(\text{vec}(\tilde{e} - e_*))\| - \epsilon C_n. \end{aligned}$$

The first result follows by rearranging. The second is an immediate implication. $\square$

B Fixed Effects in Case 1

Here we briefly discuss the asymptotic properties of the bias corrected estimator, standard error formulas, and diagnostics, when utilities depend additively on fixed effects. Recall that product and/or market fixed effects are handled by de-meaning Z_t and/or $\hat{\xi}_t$ at the product and/or market level when forming moments, as in [Somaini and Wolak \(2016\)](#) and [Conlon and Gortmaker \(2020\)](#).

Market Fixed Effects. De-meaning at the market level introduces correlation among the instruments within each market and among the $\hat{\xi}_t$ within each market. This does not affect the theoretical derivations, which already allow for arbitrary cor-

relation among the elements of Z_t and $\hat{\xi}_t$ within markets. In particular, the standard error formula (15) still applies as it clusters at the market level. The bootstrap for generating critical values for LM_2 does not need to be modified either, as it is also based on clustering at the market level.

Product Fixed Effects. De-meaning at the product level introduces correlation across markets, but this is asymptotically negligible. It is straightforward to show that the theoretical results from Section 6.1 carry over to this case. An alternative (equivalent) approach is to simply include dummy variables for each product, in which case the theory carries over directly once the parameter space is suitably expanded.

Product and Market Fixed Effects. In this case de-meaning introduces correlation both across products within markets and across markets. The correlation across products within markets is accommodated by the fact that we cluster at the market level, whereas the correlation across markets is asymptotically negligible.

C Diagnostic 2

C.1 Case 1

To introduce the diagnostic, we extend γ to $\zeta = (\gamma, \psi)$. With slight abuse of notation, we now write $\hat{\xi}_{jt}$ and m_t on this extended space as functions $\hat{\xi}_{jt}(\zeta; \eta)$ and $m_t(\zeta; \eta)$ of ζ and η , with the understanding that $\hat{\xi}_{jt}(\gamma) = \hat{\xi}_{jt}((\gamma, 0); \eta)$ and $m_t(\gamma) = m_t((\gamma, 0); \eta)$. Let

$$\hat{\lambda}(\eta) = \left(\frac{1}{T} \sum_{t=1}^T Z_t w_t(\hat{\gamma}, \eta) \right)' \hat{V}^{-1} (I - \hat{G}(\hat{G}' \hat{V}^{-1} \hat{G})^{-1} \hat{G}' \hat{V}^{-1}) \sqrt{T} \bar{g}, \quad (34)$$

$$\hat{\lambda}_t(\eta) = u_t(\hat{\gamma}, \eta)' \hat{V}_t^{-1} (I - \hat{M}_t(\hat{M}_t' \hat{V}_t^{-1} \hat{M}_t)^{-1} \hat{M}_t' \hat{V}_t^{-1}) \sqrt{T} (m_t(\hat{\gamma}) - \bar{m}_t), \quad (35)$$

where $w_t(\gamma, \eta) = (w_{jt}(\gamma, \eta))'_{j \in \mathcal{J}_t}$ is $|\mathcal{J}_t| \times \dim(\psi)$ and $u_t(\gamma, \eta)$ are $\dim(\psi) \times \dim(m)$, with

$$w_{jt}(\gamma, \eta) = \lim_{\psi \rightarrow 0} \frac{\partial \hat{\xi}_{jt}((\gamma, \psi); \eta)}{\partial \psi}, \quad j \in \mathcal{J}_t, \quad u_t(\gamma, \eta)' = \lim_{\psi \rightarrow 0} \frac{\partial m_t((\gamma, \psi); \eta)'}{\partial \psi}.$$

We take limits to deal with parameters that are at the boundary when $\psi = 0$, such as the variance of the random coefficients on η . For the intuition, the left-most terms in (34) and (35) are the Jacobian of the moments with respect to ψ . These can depend to first order on $\hat{\gamma}$. The second parts of these expressions eliminate this dependence with a similar correction to $\hat{\kappa}_{bc}$. To ensure that this elimination does not make the test statistic degenerate, in practice we form the moments and gradients in these statistics using an enlarged set of instruments formed by augmenting Z_t with an additional $\dim(\psi) \times |\mathcal{J}_t|$ set of “optimal instruments” for ψ . These are given by $\tilde{Z}_t(\eta) = \hat{w}_t(\hat{\gamma}, \eta)'$, where $\hat{w}_t$ is formed as above but using predicted prices $\hat{p}_t$ and corresponding predicted market shares $\hat{s}_t$ as in Remark 4. The dependence of $\tilde{Z}_t$ on η does not affect the properties of the statistic and we suppress this notation in what follows. We estimate the variance of $\hat{\lambda}(\eta)$ and $\hat{\lambda}_t(\eta)$ using

$$\hat{\Lambda}(\eta) = \left(\frac{1}{T} \sum_{t=1}^T Z_t w_t(\hat{\gamma}, \eta) \right)' \hat{V}^{-1} (I - \hat{G}(\hat{G}' \hat{V}^{-1} \hat{G})^{-1} \hat{G}' \hat{V}^{-1}) \left(\frac{1}{T} \sum_{t=1}^T Z_t w_t(\hat{\gamma}, \eta) \right), \quad (36)$$

$$\hat{\Lambda}_t(\eta) = u_t(\hat{\gamma}, \eta)' \hat{V}_t^{-1} (I - \hat{M}_t(\hat{M}_t' \hat{V}_t^{-1} \hat{M}_t)^{-1} \hat{M}_t' \hat{V}_t^{-1}) u_t(\hat{\gamma}, \eta).$$

Finally, define

$$\hat{W}(\eta) = \left(\hat{\lambda}(\eta) + \sum_{t=1}^{\tau} \hat{\lambda}_t(\eta) \right)' \left(\hat{\Lambda}(\eta) + \sum_{t=1}^{\tau} \hat{\Lambda}_t(\eta) \right)^{-1} \left(\hat{\lambda}(\eta) + \sum_{t=1}^{\tau} \hat{\lambda}_t(\eta) \right).$$

Without microdata, $\hat{W}(\eta)$ simplifies to $\hat{W}(\eta) = \hat{\lambda}(\eta)' \hat{\Lambda}(\eta)^{-1} \hat{\lambda}(\eta)$. The statistic $\hat{W}(\eta)$ can be shown to behave like a $\chi_{\dim(\psi)}^2$ random variable under the null (i.e., when the true $\psi = 0$), but it depends on the nuisance parameter η . Thus, we define our second diagnostic as

$$LM_2 = \sup_{\eta \in S^J: \eta \perp C(\tilde{e})} \hat{W}(\eta), \quad (37)$$

where we take the supremum over η in the unit sphere (since the scale of η is not important) that are orthogonal to the column span of $\tilde{e}$.

Following, e.g., Hansen (1996), critical values can be computed by simulation.

Draw $(\varpi_t^*)_{t=1}^T, (\varpi_{i1}^*)_{i=1}^{N_1}, \dots, (\varpi_{i\tau}^*)_{i=1}^{N_\tau}$ iid from a $N(0, 1)$ distribution, and set

$$\hat{W}^*(\eta) = \left(\hat{\lambda}^*(\eta) + \sum_{t=1}^{\tau} \hat{\lambda}_t^*(\eta) \right)' \left(\hat{\Lambda}(\eta) + \sum_{t=1}^{\tau} \hat{\Lambda}_t(\eta) \right)^{-1} \left(\hat{\lambda}^*(\eta) + \sum_{t=1}^{\tau} \hat{\lambda}_t^*(\eta) \right),$$

where

$$\begin{aligned} \hat{\lambda}^*(\eta) &= \left(\frac{1}{T} \sum_{t=1}^T Z_t w_t(\hat{\gamma}, \eta) \right)' \hat{V}^{-1} (I - \hat{G}(\hat{G}' \hat{V}^{-1} \hat{G})^{-1} \hat{G}' \hat{V}^{-1}) \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T \varpi_t^* Z_t \hat{\xi}_t(\hat{\gamma}) \right), \\ \hat{\lambda}_t^*(\eta) &= u_t(\hat{\gamma}, \eta)' \hat{V}_t^{-1} (I - \hat{M}_t(\hat{M}_t' \hat{V}_t^{-1} \hat{M}_t)^{-1} \hat{M}_t' \hat{V}_t^{-1}) \left(\frac{1}{\sqrt{T}} \sum_{i=1}^{N_t} \varpi_{it}^* (m_t(\hat{\gamma}) - m_{it}) \right). \end{aligned}$$

For each collection of $N(0, 1)$ draws, compute

$$LM_2^* = \sup_{\eta \in S^J: \eta \perp C(\bar{e})} \hat{W}^*(\eta).$$

Let $\hat{\xi}_{0.95}^*$ denote the 95th percentile of LM_2^* across a large number of independent draws. This quantity is easy to compute, as only the right-most terms in the expressions for $\hat{\lambda}^*$ and $\hat{\lambda}_t^*$ need to be recomputed for different draws and these terms do not depend on η . We reject the null that the dimension of $\tilde{e}$ is adequate if $LM_2 > \hat{\xi}_{0.95}^*$.

Example 1 (continued). Let the random coefficient on η be $\delta_i = \psi_1 + \sqrt{\psi_2} Z_i$, where $Z_i \sim F_0$ has mean zero and unit variance, $\psi_1 \in \mathbb{R}$, and $\psi_2 \in [0, \infty)$. For simplicity, suppose $\bar{x}_t$ is empty and $|\mathcal{J}_t| = J$. Define the functions $\sigma_{jt}(\cdot; (\gamma, \psi), \eta)$ and $\varsigma_{jt}(\cdot; \gamma)$ from $\mathbb{R}^J$ to $\mathbb{R}$ by

$$\begin{aligned} \sigma_{jt}(\xi_t; (\gamma, \psi), \eta) &= \int \varsigma_{jt}(\xi_t + (\psi_1 + \sqrt{\psi_2} z)\eta; \gamma) dF_0(z), \\ \varsigma_{jt}(u; \gamma) &= \int \frac{e^{-\alpha' p_{jt} + \beta' e_j + u_j}}{1 + \sum_{k \in \mathcal{J}_t} e^{-\alpha' p_{kt} + \beta' e_k + u_k}} dF(\alpha, \beta), \end{aligned}$$

where we have suppressed dependence on p_t . Here $\hat{\xi}_{jt}(\zeta; \eta)$ solves $s_{jt} = \sigma_{jt}(\hat{\xi}; \zeta, \eta)$ for $j \in \mathcal{J}_t$. Evidently, $\hat{\xi}_{jt}(\gamma) = \hat{\xi}_{jt}((\gamma, 0); \eta)$. Using $\dot{\varsigma}_{jt}(u; \gamma)$ and $\ddot{\varsigma}_{jt}(u; \gamma)$ to denote the first and second derivatives of $\varsigma_{jt}(u; \gamma)$ with respect to u , we have

$$\lim_{\psi \rightarrow 0} \frac{\partial \sigma_{jt}(\hat{\xi}_t((\gamma, \psi); \eta); (\gamma, \psi), \eta)}{\partial \psi} = \begin{pmatrix} \eta' \dot{\varsigma}_{jt}(\hat{\xi}_t(\gamma); \gamma) \\ \frac{1}{2} \eta' \ddot{\varsigma}_{jt}(\hat{\xi}_t(\gamma); \gamma) \eta \end{pmatrix}.$$

It follows by the implicit function theorem that

$$w_t(\gamma, \eta) = \begin{pmatrix} -\eta & -\left(\frac{\partial \varsigma_t(\hat{\xi}_t(\gamma); \gamma)}{\partial \xi'}\right)^{-1} \left(\frac{1}{2} \eta' \ddot{\varsigma}_{jt}(\hat{\xi}_t(\gamma); \gamma) \eta\right)_{j \in \mathcal{J}_t} \end{pmatrix},$$

where $\varsigma_t(u; \gamma) = (\varsigma_{jt}(u; \gamma))_{j \in \mathcal{J}_t}$. Plugging this into (34) and (36), one can compute $\hat{W}(\eta)$ and thus the LM_2 diagnostic.

C.2 Case 2

We augment $\tilde{e}$ with a vector η representing additional attributes not included in $\tilde{e}$ and augment θ with an additional component $\psi \in \Psi$ representing coefficients on η . Correspondingly, we extend γ to $\zeta = (\gamma, \psi)$. With slight abuse of notation, we now write choice probabilities on this extended space as $\sigma_{ij}(\zeta; \eta)$, with the understanding that $\sigma_{ij}(\gamma) = \sigma_{ij}((\gamma, 0); \eta)$.

Consider an LM test of the null hypothesis that $\psi = 0$. Such a test could be based on the score

$$\frac{1}{n} \sum_{i=1}^n \sum_{j=0}^J \frac{d_{ij}}{\sigma_{ij}(\hat{\gamma})} w_{ij}(\hat{\gamma}, \eta), \quad (38)$$

where

$$w_{ij}(\gamma, \eta) = \lim_{\psi \rightarrow 0} \frac{\partial \sigma_{ij}((\gamma, \psi); \eta)}{\partial \psi}.$$

Example 3 (continued). Let the random coefficient on η be $\delta_i \sim \psi Z_i$, where $\psi \in [0, \infty)$ and $Z_i \sim F_0$ has mean zero and unit variance. For simplicity, suppose $\Pi = 0$. Define the functions $\sigma_{ij}(\cdot; (\gamma, \psi), \eta)$ and $\varsigma_{ij}(\cdot; (\gamma, \psi), \eta)$ from $\mathbb{R}^J$ to $\mathbb{R}$ by

$$\begin{aligned} \sigma_{ij}((\gamma, \psi); \eta) &= \int \varsigma_{ij}(\xi + \sqrt{\psi} \eta z; \gamma) dF_0(z), \\ \varsigma_{ij}(u; \gamma) &= \int \frac{e^{\alpha' p_{ij} + \beta' e_j + u_j}}{1 + \sum_{k=1}^J e^{\alpha' p_{ik} + \beta' e_k + u_k}} dF(\alpha, \beta), \end{aligned}$$

where we have suppressed dependence on p_{ij} and $\bar{x}$. Using $\dot{\varsigma}_{ij}$ and $\ddot{\varsigma}_{ij}$ to denote first and second derivatives of ς_{ij} with respect to its first argument, we have

$$\frac{\partial \sigma_{ij}((\gamma, \psi); \eta)}{\partial \psi} = \frac{1}{2\sqrt{\psi}} \int z \eta' \dot{\varsigma}_{ij}(\xi + \sqrt{\psi} \eta z; \gamma) dF_0(z).$$

Thus, $w_{ij}(\gamma, \eta) = \frac{1}{2}\eta' \ddot{\zeta}_{ij}(\xi; \gamma)\eta$. The term $w_{ij}(\hat{\gamma}, \eta) = \frac{1}{2}\eta' \ddot{\zeta}_{ij}(\hat{\xi}; \hat{\gamma})\eta$ is plugged into the LM_2 statistic below.

The statistic (38) can still depend to first order on $\hat{\gamma}$. To eliminate this dependence, we perform a similar correction to $\hat{\kappa}_{bc}$. Let $g_i(\gamma; \eta) = w_i(\gamma, \eta)'V_i(\gamma)^{-1}(d_i - \sigma_i(\gamma))$, where $w_i(\gamma; \eta) = (w_{ij}(\gamma; \eta))_{j=1}^J$ is $J \times \dim(\psi)$. Then define the $\dim(\psi) \times J$ matrix

$$\hat{c}_i(\eta)' = \left(w_i(\hat{\gamma}, \eta) + \dot{\sigma}_i(\hat{\gamma})\hat{H}^{-1} \left(\frac{1}{n} \sum_{l=1}^n \dot{g}_l(\hat{\gamma}; \eta)' \right) \right)' V_i(\hat{\gamma})^{-1},$$

where $\dot{g}_i(\gamma; \eta)' = \frac{\partial g_i(\gamma; \eta)}{\partial \gamma}$ is $\dim(\gamma) \times \dim(\psi)$, and let

$$\hat{\lambda}(\eta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \hat{c}_i(\eta)'(d_i - \sigma_i(\hat{\gamma})), \quad \hat{\Lambda}(\eta) = \frac{1}{n} \sum_{i=1}^n \hat{c}_i(\eta)'V_i(\hat{\gamma})\hat{c}_i(\eta).$$

Finally, let

$$\hat{W}(\eta) = \hat{\lambda}(\eta)' \hat{\Lambda}(\eta)^{-1} \hat{\lambda}(\eta).$$

It can be shown that the statistic $\hat{W}(\eta)$ behaves like a $\chi_{\dim(\psi)}^2$ random variable under the null (i.e., when the true $\psi = 0$), but it depends on the nuisance parameter η . Thus, we again define our second diagnostic as

$$LM_2 = \sup_{\eta \in S^J: \eta \perp C(\bar{e})} \hat{W}(\eta). \quad (39)$$

Critical values can be computed by simulation: for iid $N(0, 1)$ random variables $(\varpi_i^*)_{i=1}^n$, compute

$$LM_2^* = \sup_{\eta \in S^J: \eta \perp C(\bar{e})} \hat{W}^*(\eta),$$

where $\hat{W}^*(\eta) = \hat{\lambda}^*(\eta)' \hat{\Lambda}(\eta)^{-1} \hat{\lambda}^*(\eta)$, with $\hat{\lambda}^*(\eta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varpi_i^* \hat{c}_i(\eta)'(d_i - \sigma_i(\hat{\gamma}))$. Note $\hat{\lambda}^*(\eta)$ factors into the product of terms involving η , which only need to be computed once, and $(\varpi_i^*)_{i=1}^n$, making it trivial to compute. Let $\hat{\xi}_{0.95}^*$ denote the 95th percentile of LM_2^* across a large number of independent sequences $(\varpi_i^*)_{i=1}^n$. The null that the dimension of $\bar{e}$ is adequate can be rejected if $LM_2 > \hat{\xi}_{0.95}^*$.

D Additional Details for Section 4.1

In this section we give additional details on the simulation design used to generate results in Section 4.1 and present additional results not included in the main text.

D.1 Simulation Design

We first fit a mixed logit model to the data using PyBLP. We use the third model in the PyBLP tutorial. The model has product fixed effects, and random coefficients on price, sugar content, mushy, and a constant for the inside products, with a diagonal covariance matrix. In addition to the 20 instruments provided in the PyBLP data, we estimate the model using two additional differentiation IVs, following [Gandhi and Houde \(2019\)](#). For product j in market t , we form

$$\sum_{j' \neq j} \mathbb{I}[|d_{j,j'}| < \hat{\sigma}_j](\hat{p}_{jt} - \hat{p}_{j't}), \quad \sum_{j' \neq j} \mathbb{I}[|d_{j,j'}| < \hat{\sigma}_j](\hat{p}_{jt} - \hat{p}_{j't})^2, \quad (40)$$

where $d_{j,j'}$ is the difference between the sugar content of focal product j and other product j' , $\hat{\sigma}_j$ is the standard deviation of $|d_{j,j'}|$ across products j' , and $\hat{p}_{jt}$ are the fitted values from a regression of p_{jt} on z_t across markets (i.e., regression coefficients are estimated at the product level).

The estimated coefficients are $\bar{\alpha} = -30.1587$, $\sigma_\alpha = 0.1431$, $\sigma_{\text{inside}} = 0.0303$, $\sigma_{\text{sugar}} = 0.0365$, and $\sigma_e = 0.5272$. The parameter σ_e is estimated imprecisely, with a standard error of 1.51. To make the simulations more challenging, we increase σ_e to 2.0, which is well inside the 95% confidence interval for σ_e based on the PyBLP estimates. We use these parameter values to generate choices given prices and exogenous variables in the simulations.

Let $\mu_{z,j}$ denote the sample mean of z_{jt} across markets, and Σ_z and Σ_ξ denote the sample variance of $z_{jt} - \mu_{z,j}$ and $\hat{\xi}_{jt}$, respectively, across products and markets, where $\hat{\xi}_{jt}$ are the estimated residuals at the data-generating parameters. For each simulated data set, we first draw ε_{jt}^* iid $N(0, \Sigma_z)$ and f_{0t}^* and f_{1t}^* iid $N(0, 1)$, and set $z_{jt}^* = \mu_{z,j} + \sqrt{1 - \rho_z} \varepsilon_{jt}^* + \sqrt{\rho_z} \text{diag}(\Sigma_z) f_{e_{jt}}^*$ with $\rho_z = 0.8$. This ensures that the mean and variances of the 20 instruments z_{jt}^* match those in the data, but with a factor structure that makes the instruments more informative about the nonlinear parameters. We also draw ξ_{jt}^* iid $N(0, \Sigma_\xi)$. To generate prices, we first run a pooled

regression of prices p_{jt} on z_{jt} , $\hat{\xi}_{jt}$, and product fixed effects. We then set $p_{jt}^* = \hat{p}_{jt}^* + u_{jt}^*$, where $\hat{p}_{jt}$ denotes the predicted values from this regression at $z_{jt} = z_{jt}^*$ and $\hat{\xi}_{jt} = \xi_{jt}^*$, and u_{jt}^* is an iid $N(0, \sigma_p^2)$ error term, where σ_p^2 is the sample variance of the residuals from the pooled price regression. Given the simulated $p_t^* = (p_{jt}^*)_{j=1}^J$ and $\xi_t^* = (\xi_{jt}^*)_{j=1}^J$, we then simulate market shares at the above parameter values (with product fixed effects also matching parameter estimates). Finally, for each j and t we form two more differentiation IVs as in (40). This yields a total of 22 instruments per product, which we use to estimate the parameters and implement our bias corrections.

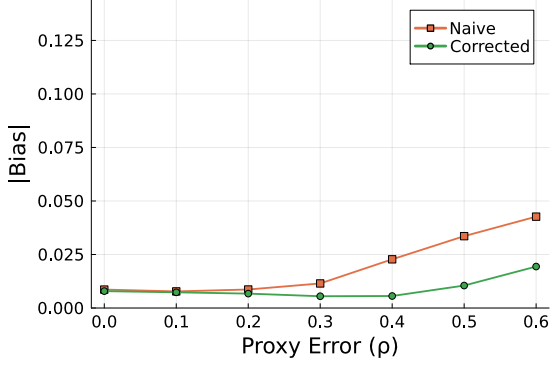
D.2 Additional Results with (Potentially) Mismeasured IVs

Here we present simulation results using two additional differentiation IVs based on the (potentially) mismeasured product attributes. In addition to the 22 instruments described above, we use two more based on the embeddings:

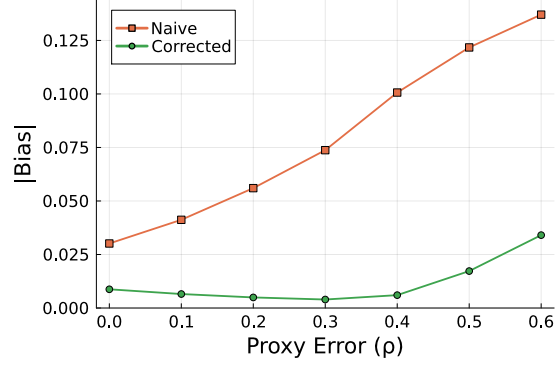
$$\sum_{j' \neq j} \mathbb{I}[d_{j,j'}^e = 0](\hat{p}_{jt} - \hat{p}_{j't}), \quad \sum_{j' \neq j} \mathbb{I}[d_{j,j'}^e = 0](\hat{p}_{jt} - \hat{p}_{j't})^2,$$

where $d_{j,j'}^e = \mathbb{I}[\tilde{e}_j \leq 0.5] - \mathbb{I}[\tilde{e}_{j'} \leq 0.5]$. When there is no proxy error ($\rho = 0$), these capture differentiation in price for products that are close in “mushiness.” However, for $\rho > 0$ these capture differentiation in price for products that are close according to the imperfect proxy $\tilde{e}_j$. Bias and RMSEs are presented in Figure 7, and closely match those in Figure 1 (which excludes the last two differentiation IVs based on proximity in $\tilde{e}_j$). Coverage of 95% confidence intervals for the counterfactuals based on the bias-corrected estimator and the variance formula (16) are presented in Table 3. These are close to the values reported in Table 1 in the main text.

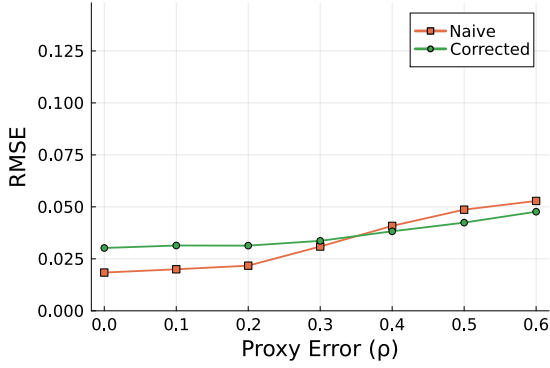
Figure 7: Bias and RMSE as a function of proxy error, [Nevo \(2001\)](#) design



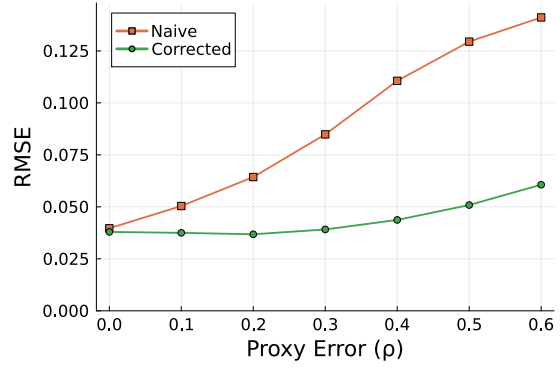
(a) Bias for the “all” counterfactual



(b) Bias for the “mushy” counterfactual



(c) RMSE for the “all” counterfactual



(d) RMSE for the “mushy” counterfactual

Table 3: Coverage of 95% confidence intervals for counterfactuals, [Nevo \(2001\)](#) design

ρ	Coverage: “all”	Coverage: “mushy”
0.00	0.958	0.962
0.10	0.966	0.971
0.20	0.966	0.972
0.30	0.959	0.967
0.40	0.953	0.940
0.50	0.923	0.870
0.60	0.899	0.769

E Additional Results for Section 5

Figure 8 compares the hit rates with and without bias correction for additional unstructured data sources and ML models. For book titles and descriptions, we use the same ML models used for book reviews (discussed in Section 5). For images, we use standard pre-trained classification models (see CMS for more details).

For 14 out of 16 proxies (around 88%), the bias correction weakly improves performance and the magnitude of the improvement is large in several cases. This confirms that the bias correction is helpful in improving the estimator’s performance at counterfactual predictions across various unstructured data sources.

F Reparameterization for Micro BLP

Consider Example 2. We partition $\beta_i = (\beta_{\bar{x},i}, \beta_{e,i})$ and $\Pi = [\Pi_{\bar{x}} \ \Pi_e \ \Pi_p]$, and write the utilities as:

$$u_{ijt} = \beta'_{\bar{x},i} \bar{x}_{jt} + \beta'_{e,i} e_j - \alpha_i p_{jt} + (\bar{x}_{jt}, p_{jt})' [\Pi_{\bar{x}} \ \Pi_p] y_{it} + e'_j \Pi_e y_{it} + \pi' \bar{y}_{ijt} + \xi_{jt} + \varepsilon_{ijt}, \quad j \in \mathcal{J}_t.$$

Suppose $\alpha_i \sim N(\bar{\alpha}, \sigma_\alpha^2)$, $\beta_{\bar{x},i} \sim N(\bar{\beta}_{\bar{x}}, \Sigma_{\bar{x}})$, $\beta_{e,i} \sim N(\bar{\beta}_e, \Sigma_e)$, and α_i , $\beta_{\bar{x},i}$ and $\beta_{e,i}$ are independent. Then,

$$\theta = (\bar{\alpha}, \sigma_\alpha, \bar{\beta}_{\bar{x}}, \bar{\beta}_e, \pi, v(\Pi_{\bar{x}}), v(\Pi_p), v(\Pi_e), l(\Sigma_{\bar{x}}), l(\Sigma_e)),$$

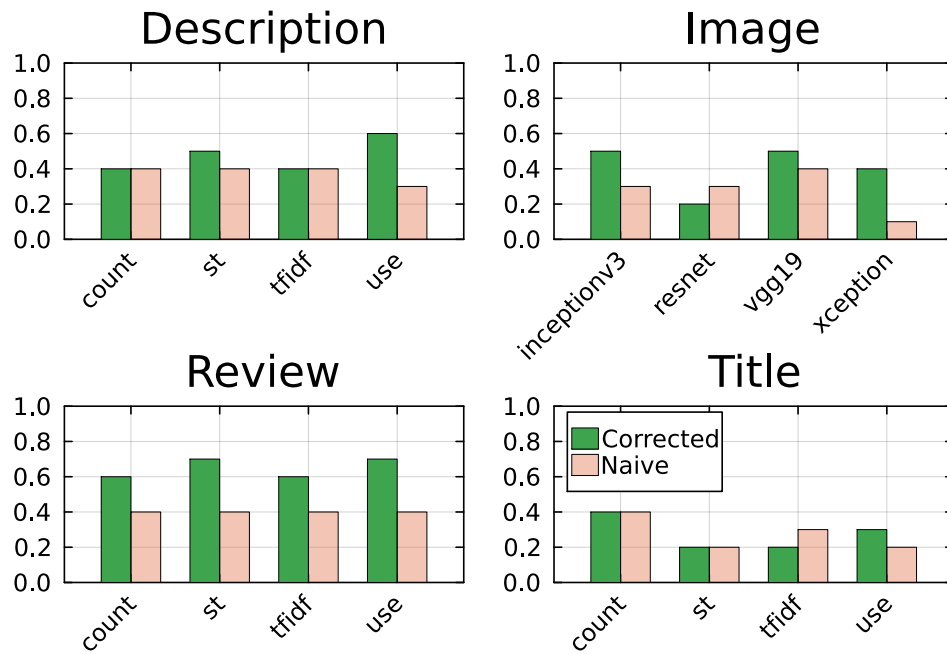
where we use the same notation v and l as for Examples 1 and 3.²³ Note that e_j only enters via $\beta'_{e,i} e_j$ and $e'_j \Pi_e y_{it}$. As before, collecting $\beta'_{e,i} e_j$ across products, we have $e \beta_{e,i} \sim N(e \bar{\beta}_e, e \Sigma_e e')$, where $e \Sigma_e e'$ has rank $r \leq J$ because e is $J \times r$. Hence,

$$\gamma(\theta, e) = (\bar{\alpha}, \sigma_\alpha, \bar{\beta}_{\bar{x}}, e \bar{\beta}_e, \pi, v(\Pi_{\bar{x}}), v(\Pi_p), v(e \Pi_e), l(\Sigma_{\bar{x}}), l_r(e \Sigma_e e')),$$

with l_r as in Examples 1 and 3.

²³As with Example 1, if $\Sigma_{\bar{x}}$ and/or Σ_e are diagonal, then we replace $l(\Sigma_{\bar{x}})$ and/or $l(\Sigma_e)$ with vectors containing their diagonal entries.

Figure 8: Rates of correct closest-substitutes predictions.



Note: Darker green bars show the fraction of books for which the bias-corrected estimator correctly identifies the closest substitute. Lighter orange bars show the corresponding fractions for the naive estimator. Each of the four panel corresponds to an unstructured data source. Within each panel, each pair of bars corresponds to an ML model used to extract the proxies from that data source.